

Lecture 14

Work and energy principle in planar motion

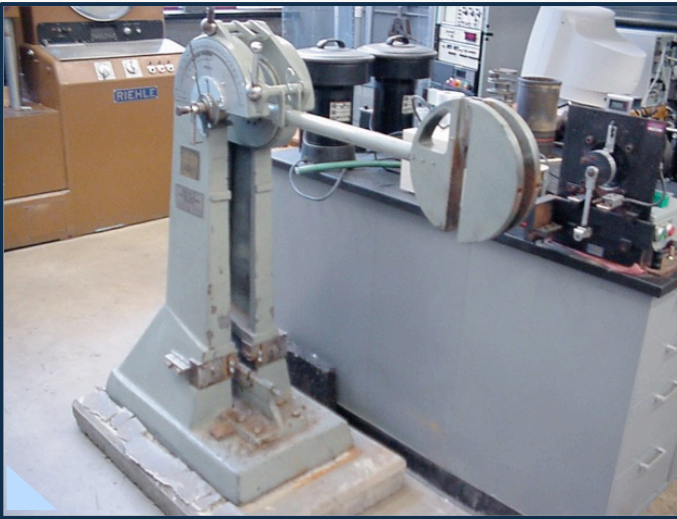
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MC2103
Fall 2026



14 Work and energy principle in planar motion

Introduction

A Charpy impact test is used to determine the amount of energy absorbed by a material during impact.



To determine the amount of energy absorbed, the final gravitational potential energy of the arm is subtracted from its initial gravitational potential energy.

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Introduction

**Method of
work and
energy**

**Method of
impulse and
momentum**

- ➡ Two methods will be used to analyze the plane motion of rigid bodies and systems of rigid bodies.

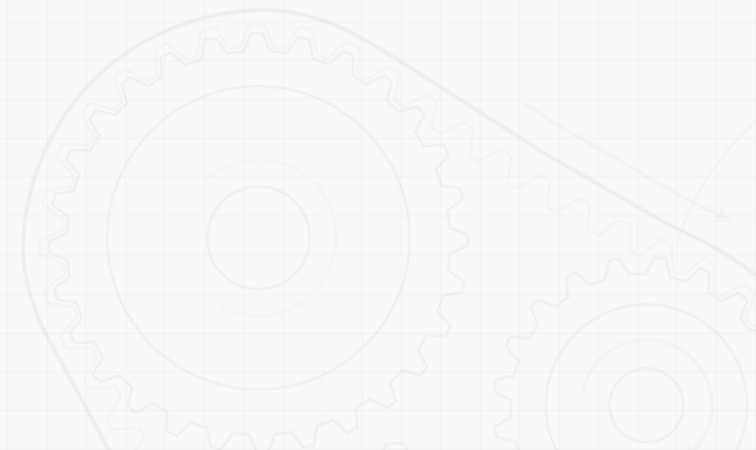
14 Work and energy principle in planar motion

Introduction

⚙ Principle of work and energy

- Principle of work and energy is well suited to the solution of problems involving displacements and velocities.

$$T_1 + U_{1 \rightarrow 2} = T_2$$



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Introduction

⚙ Principle of impulse and momentum

- Principle of impulse and momentum is appropriate for problems involving velocities and time.

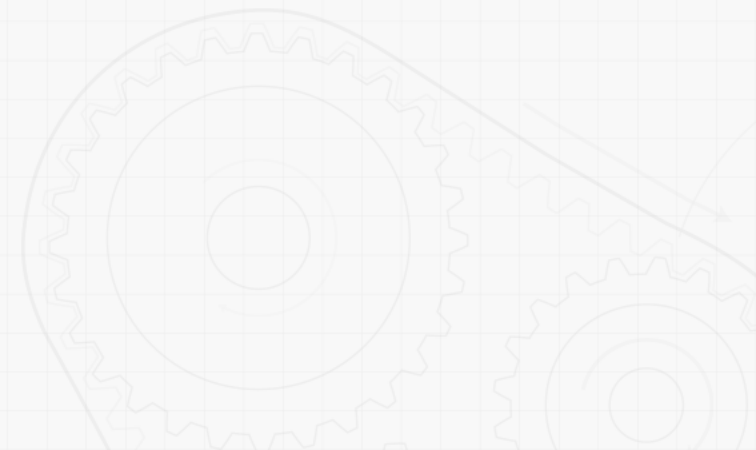
$$\vec{L}_1 + \sum \int_{t_1}^{t_2} \vec{F} dt = \vec{L}_2 \quad (\vec{H}_O)_1 + \sum \int_{t_1}^{t_2} \vec{M}_O dt = (\vec{H}_O)_2$$

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Introduction

⚙️ Solving Problems involving eccentric impact

- Problems involving eccentric impact are solved by supplementing the principle of impulse and momentum with the application of the coefficient of restitution.



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Introduction

⚙ Approaches to Rigid Body Kinetics Problem

Forces and
Accelerations



Newton's Second
Law(last chapter)

$$\sum \vec{F} = m\vec{a}_G$$
$$\sum \vec{M}_G = \dot{\vec{H}}_G$$

Velocities and
Displacements



Work-Energy

$$T_1 + U_{1 \rightarrow 2} = T_2$$

Velocities and
Time



Impulse-
Momentum

$$m\bar{v}_1 + \int_{t_1}^{t_2} \vec{F} dt = m\bar{v}_2$$
$$I_G\omega_1 + \int_{t_1}^{t_2} M_G dt = I_G\omega_2$$

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Principle of Work and Energy

⚙ **Work and kinetic energy are scalar quantities.**

- Assume that the rigid body is made of a large number of particles.

$$T_1 + U_{1 \rightarrow 2} = T_2$$

 T_1, T_2

initial and final total kinetic energy of particles forming body

 U_1, U_2

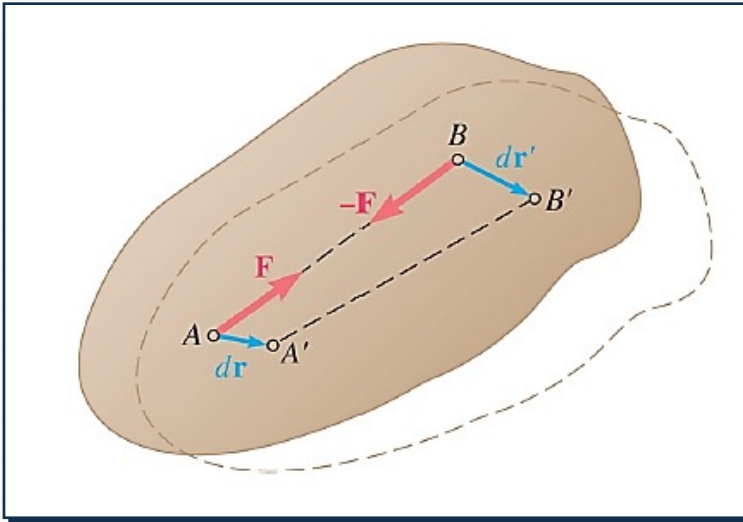
total work of internal and external forces acting on particles of body

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Principle of Work and Energy

⚙ **Work and kinetic energy are scalar quantities.**

- Assume that the rigid body is made of a large number of particles.



- Internal forces between particles A and B are equal and opposite.

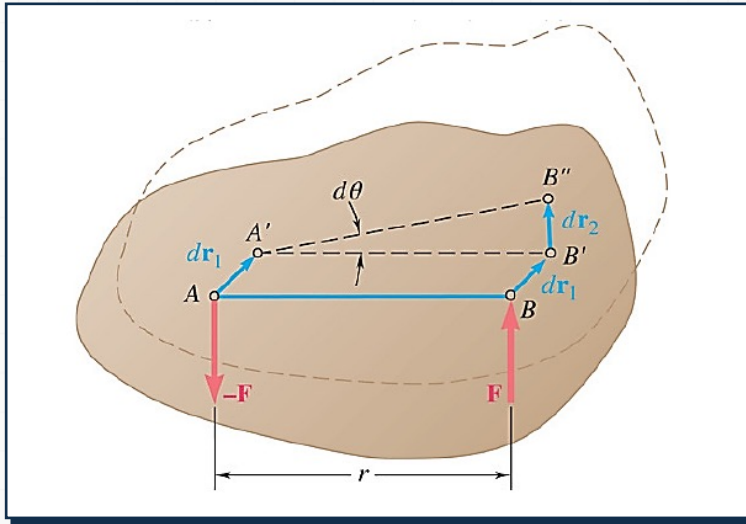
➡ Therefore, **the net work of internal forces is zero.**

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Work of Forces Acting on a Rigid Body

⚙ Recall that work is force times displacement along its path.

- Work of a force during a displacement of its point of application,

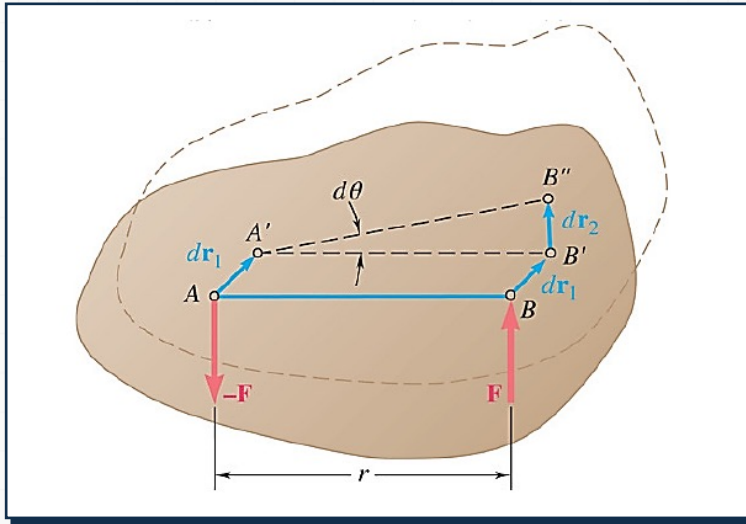


$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} \vec{F} \cdot d\vec{r} = \int_{S_1}^{S_2} (F \cos \alpha) ds$$

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Work of Forces Acting on a Rigid Body

⚙️ **Two equal and opposite forces forming a couple do work.**



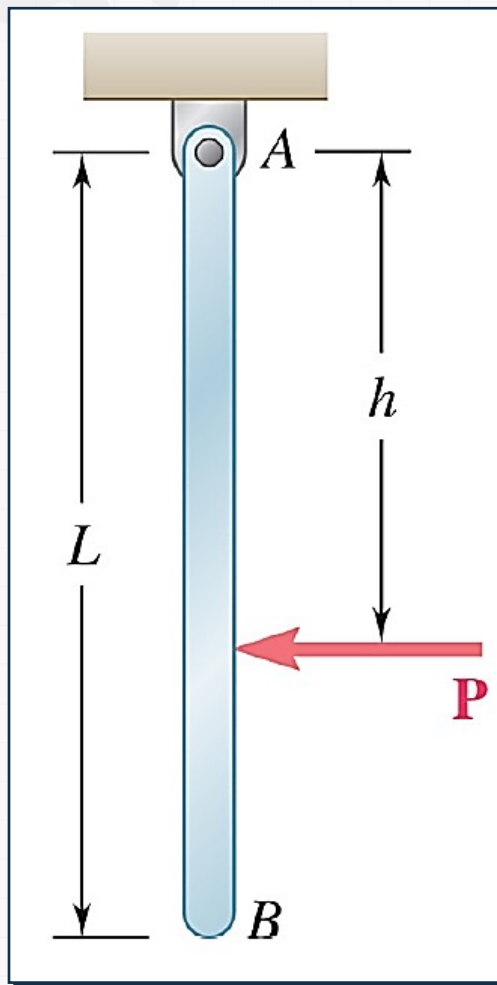
- Consider the net work of two forces $\vec{F}$ and $-\vec{F}$ forming a couple of moment $\vec{M}$ during a displacement of their points of application.

$$\begin{aligned} dU &= \vec{F} \cdot d\vec{r}_1 - \vec{F} \cdot d\vec{r}_1 + \vec{F} \cdot d\vec{r}_2 \\ &= F ds_2 = Fr d\theta \\ &= M d\theta \end{aligned}$$

$$U_{1 \rightarrow 2} = \int_{\theta_1}^{\theta_2} M d\theta = M(\theta_2 - \theta_1) \text{ if } M \text{ is constant.}$$

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Work of Forces Acting on a Rigid Body



⚙ Do the pin forces at point A do work?

a) Yes

b) No

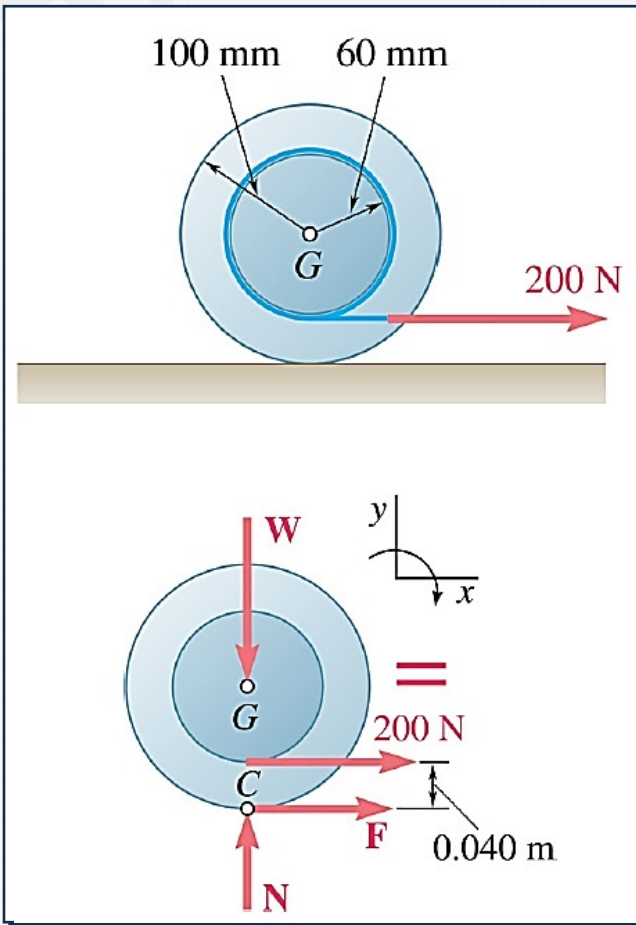
⚙ Does the force P do work?

a) Yes

b) No

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Work of Forces Acting on a Rigid Body



⚙ Does the normal force N do work on the disk?

a) Yes

b) No

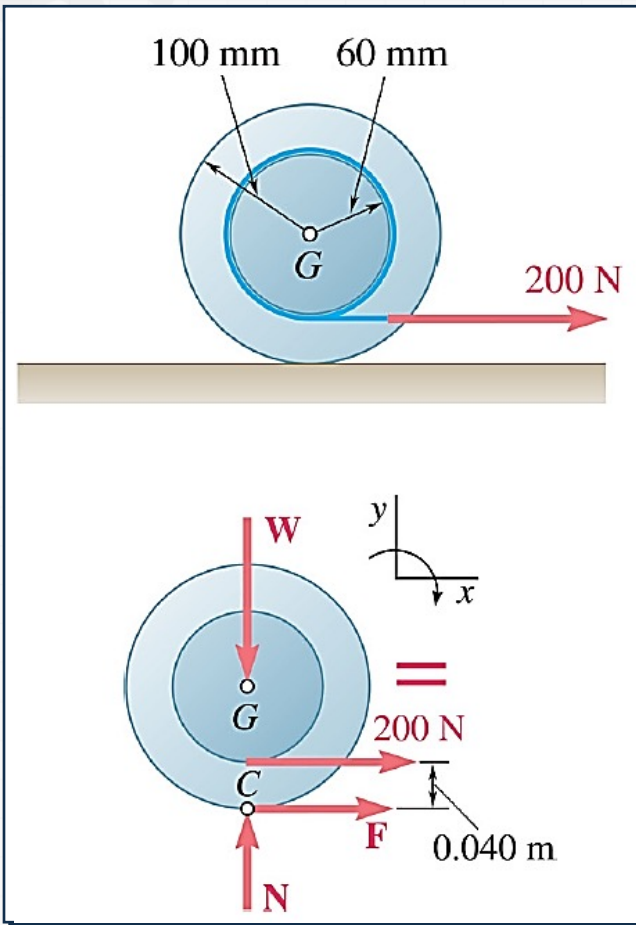
⚙ Does the weight W do work?

a) Yes

b) No

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Work of Forces Acting on a Rigid Body



⚙ If the disk rolls without slip, does the friction force F do work?

a) Yes

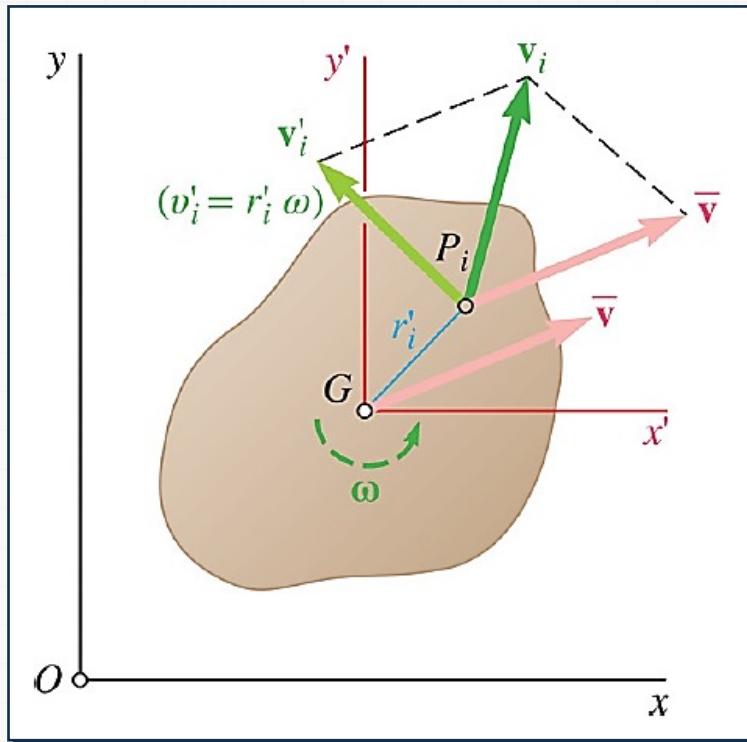
b) No

$$dU = F ds_C = F(v_2 dt) = 0$$

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Kinetic Energy of a Rigid Body in Plane Motion

⚙️ Consider a rigid body of mass m in plane motion consisting of individual particles i .



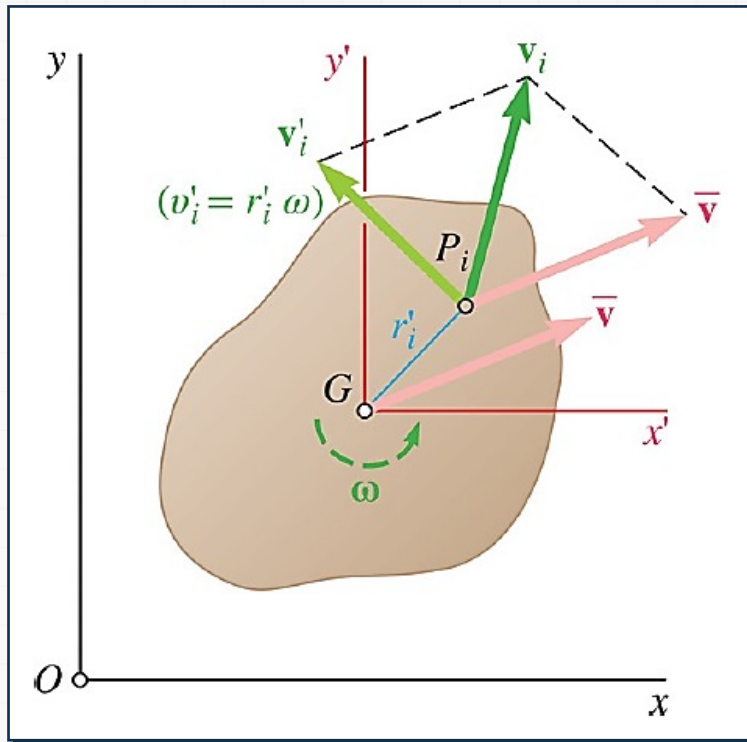
• The kinetic energy of the body can then be expressed as

$$\begin{aligned} T &= \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \sum \Delta m_i v_i'^2 \\ &= \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \left(\sum r_i'^2 \Delta m_i \right) \omega^2 \\ &= \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \bar{I} \omega^2 \end{aligned}$$

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Kinetic Energy of a Rigid Body in Plane Motion

⚙️ Consider a rigid body of mass m in plane motion consisting of individual particles i .



• Kinetic energy of a rigid body can be separated into

- The kinetic energy associated with the motion of the mass center G .
- The kinetic energy associated with the rotation of the body about G .

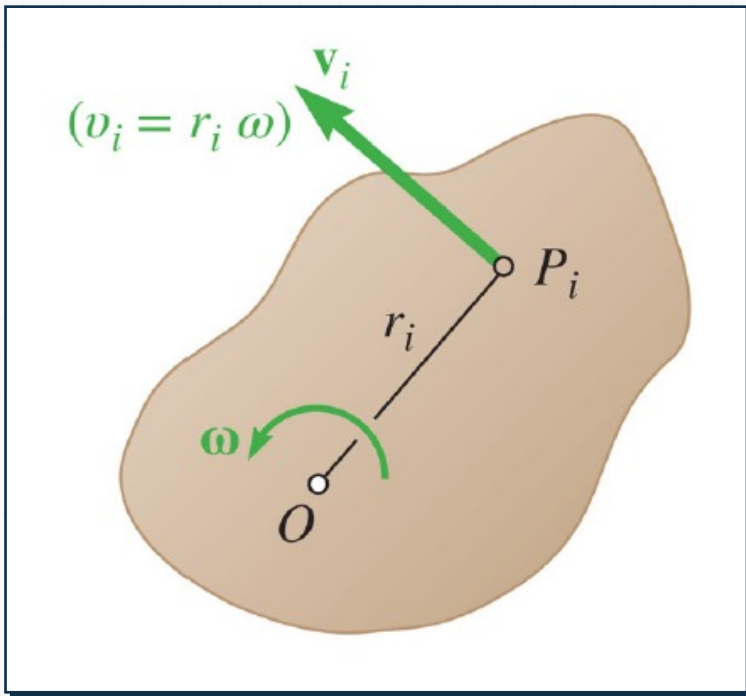
$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\bar{I}\omega^2$$

Translation + Rotation

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Kinetic Energy of a Rigid Body in Plane Motion

⚙️ Consider a rigid body rotating about a fixed axis through O .

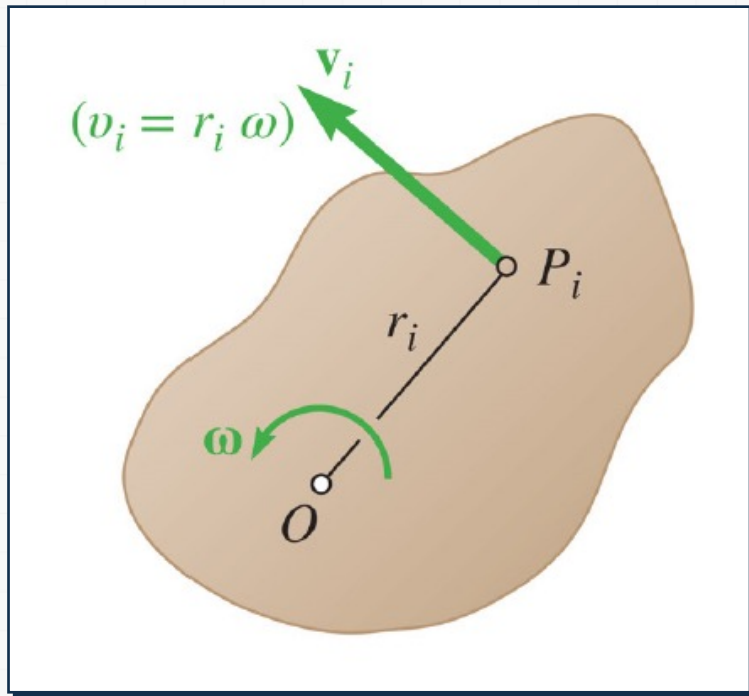


$$T = \frac{1}{2} \sum \Delta m_i v_i^2 + \frac{1}{2} \sum \Delta m_i (r_i \omega)^2 = \frac{1}{2} \left(\sum r_i^2 \Delta m_i \right) \omega^2$$
$$= \frac{1}{2} I_O \omega^2$$

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Kinetic Energy of a Rigid Body in Plane Motion

⚙️ Consider a rigid body rotating about a fixed axis through O .



- This is equivalent to using

$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\bar{I}_G\omega^2$$

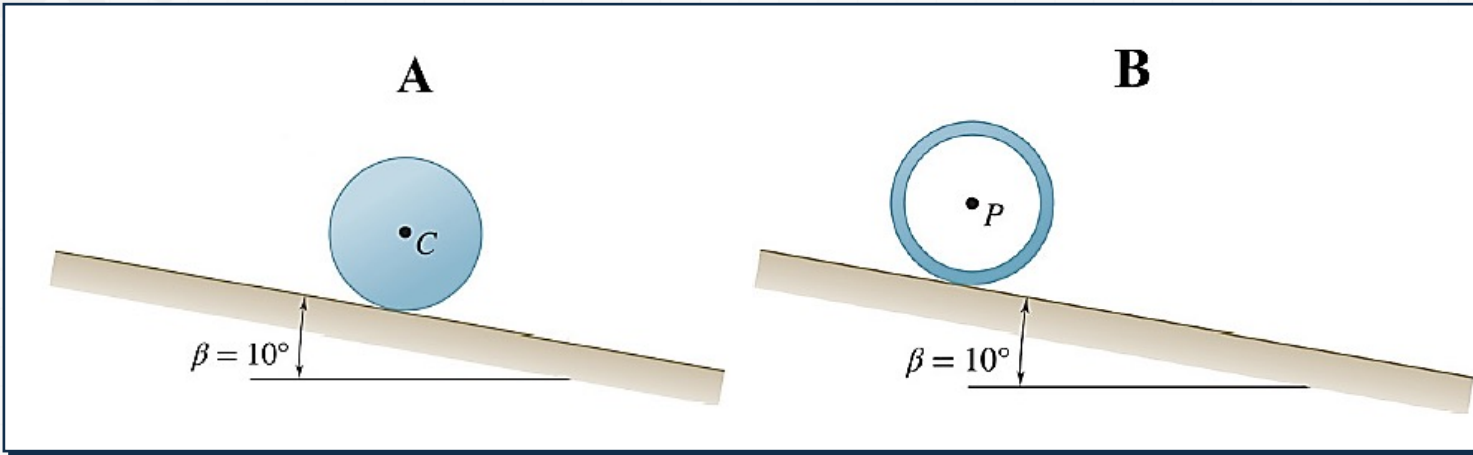
- Remember to only use

$$T = \frac{1}{2}I_O\omega^2$$

when O is a fixed axis of rotation.

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Concept Quiz

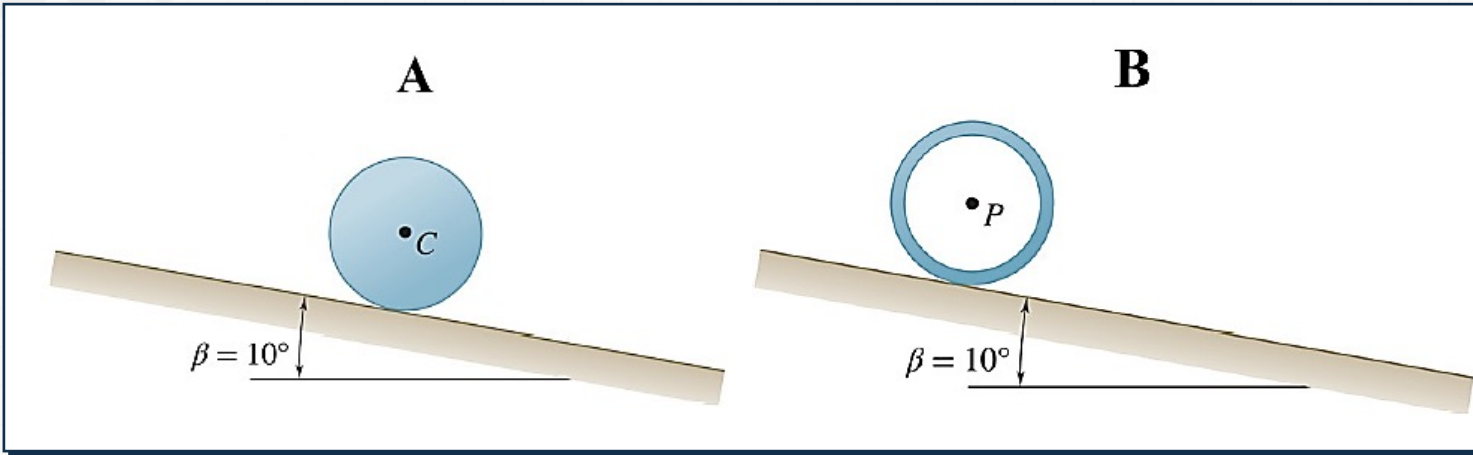


⚙ The solid cylinder A and the pipe B have the same diameter and mass. If they are both released from rest at the top of the hill, which will reach the bottom the fastest?

- a) A will reach the bottom first
- b) B will reach the bottom first
- c) They will reach the bottom at the same time

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Concept Quiz



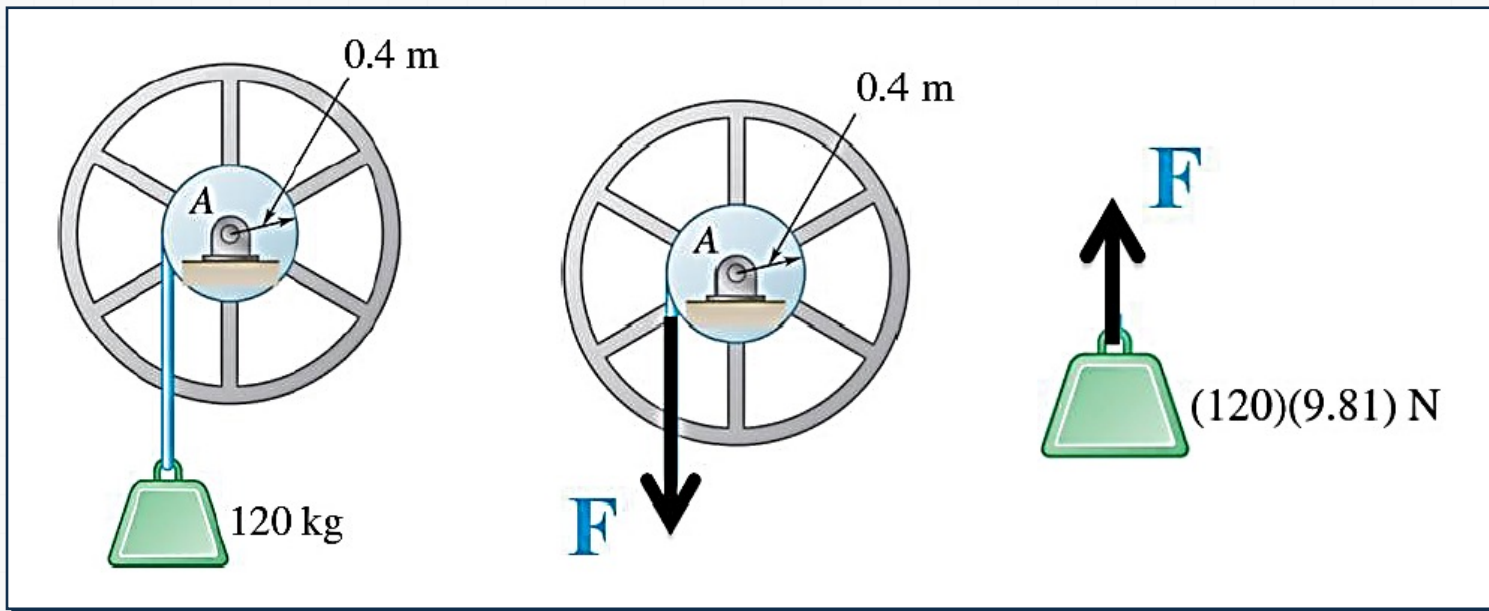
⚙ Which will have the greatest kinetic energy when it reaches the bottom?

- a) Cylinder A b) Pipe B c) Same kinetic energy

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Systems of Rigid Bodies

For problems involving systems consisting of several rigid bodies, the principle of work and energy can be applied to each body.



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Systems of Rigid Bodies

⚙️ We may also apply the principle of work and energy to the entire system,

$$T_1 + U_{1 \rightarrow 2} = T_2$$

 T_1, T_2

arithmetic sum of the kinetic energies of all bodies forming the system

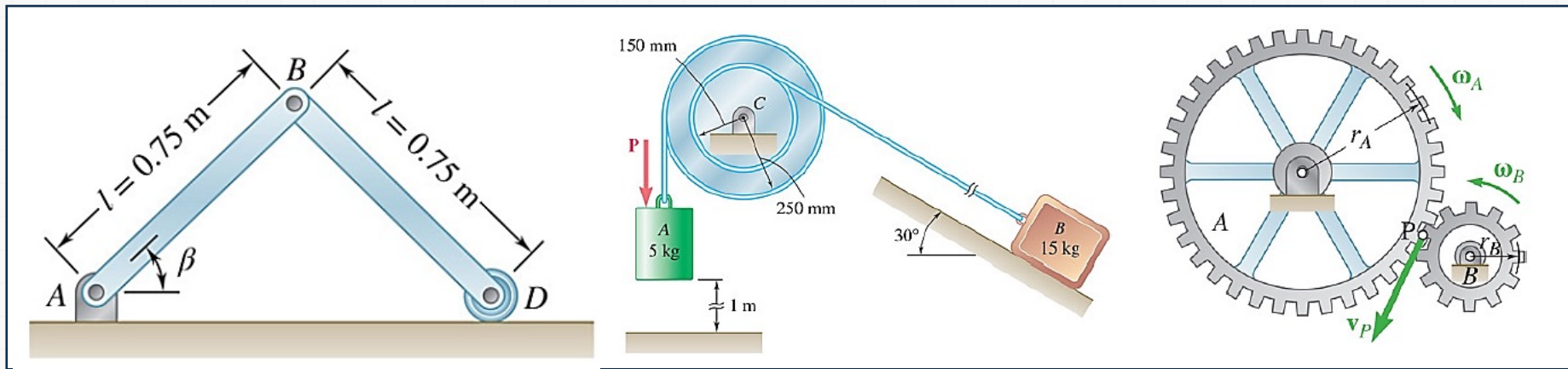
 $U_{1 \rightarrow 2}$

work of all forces acting on the various bodies, whether these forces are internal or external to the system as a whole.

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Systems of Rigid Bodies

- ⚙️ **For problems involving pin connected members, blocks and pulleys connected by inextensible cords, and meshed gears,**
 - Internal forces occur in pairs of equal and opposite forces
 - Points of application of each pair move through equal distances
 - Net work of the internal forces is zero
 - Work on the system reduces to the work of the external forces



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Conservation of Energy

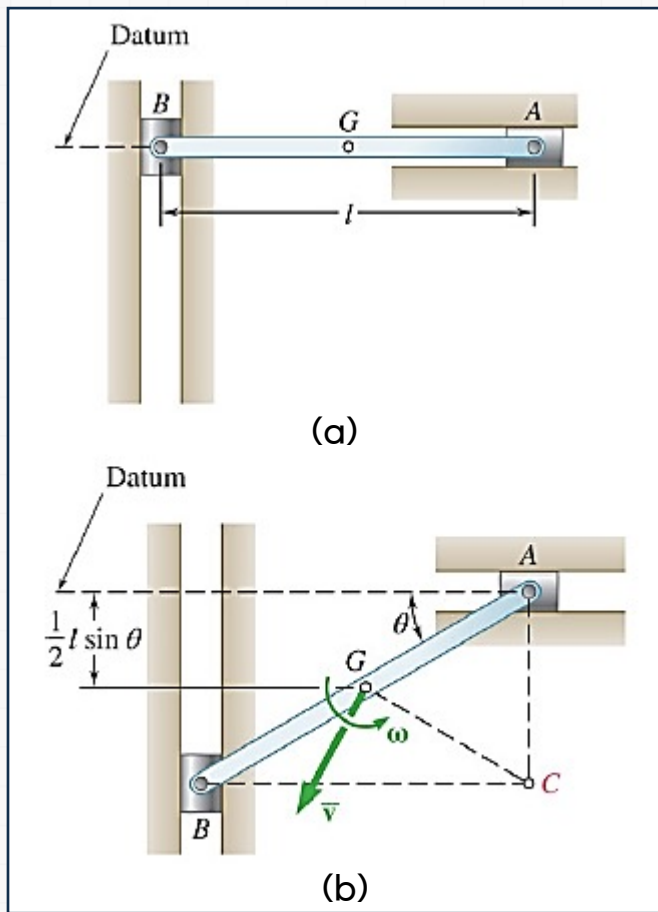
Expressing the work of conservative forces as a change in potential energy, the principle of work and energy becomes

$$T_1 + V_1 = T_2 + V_2$$

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Conservation of Energy

⚙ Consider the slender rod of mass m .

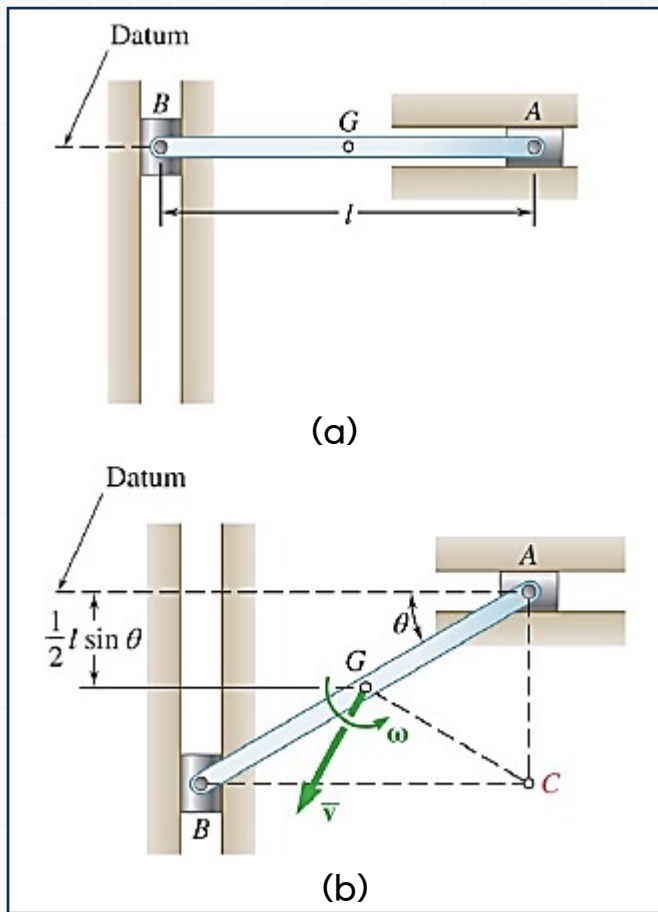


- mass m
- released with zero velocity
- determine w at q

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Conservation of Energy

⚙ Consider the slender rod of mass m .



$$T_1 = 0, \quad V_1 = 0$$

$$T_2 = \frac{1}{2}m\bar{v}_2^2 + \frac{1}{2}\bar{I}\omega_2^2$$

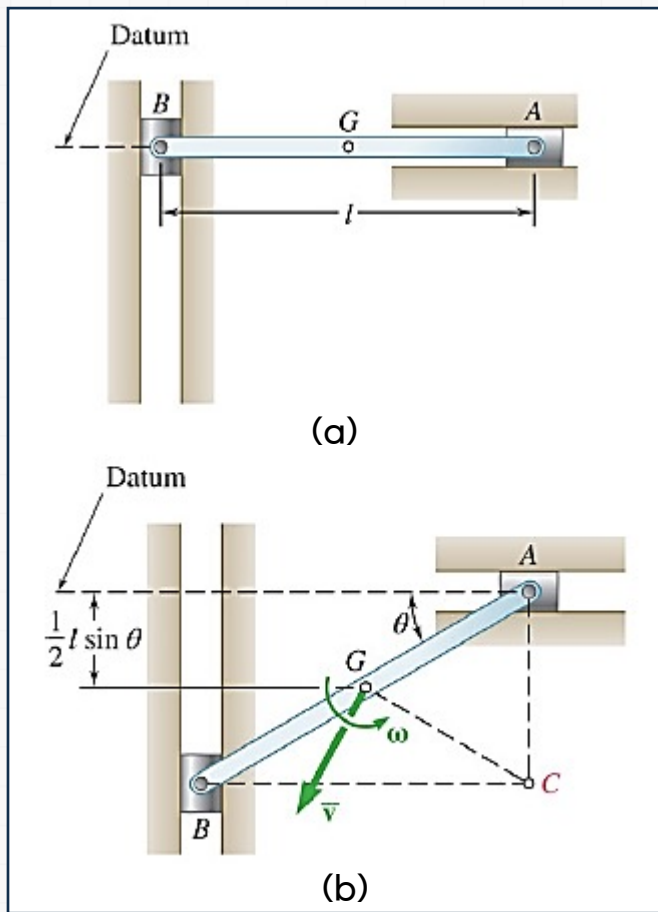
$$= \frac{1}{2}m\left(\frac{1}{2}l\omega\right)^2 + \frac{1}{2}\left(\frac{1}{12}ml^2\right)\omega^2 = \frac{1}{2}\frac{ml^2}{3}\omega^2$$

$$V_2 = -\frac{1}{2}Wl \sin \theta = -\frac{1}{2}mgl \sin \theta$$

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Conservation of Energy

⚙️ Consider the slender rod of mass m .



$$T_1 + V_1 = T_2 + V_2$$

$$0 = \frac{1}{2} \frac{ml^2}{3} \omega^2 - \frac{1}{2} mgl \sin \theta$$

$$\omega = \sqrt{\frac{3g}{l} \sin \theta}$$

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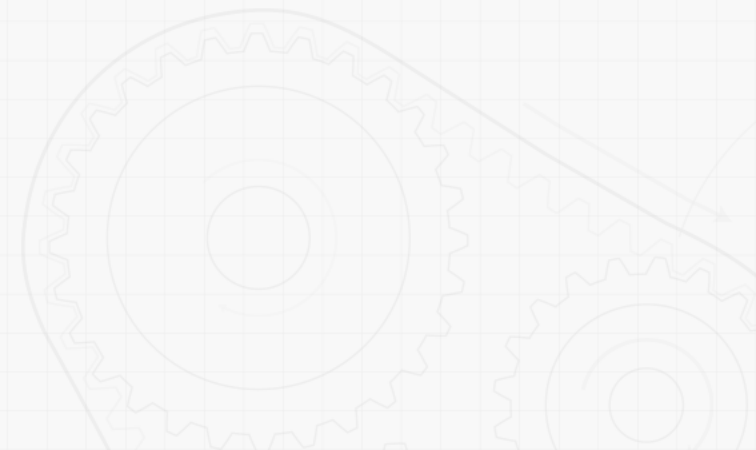
Power

⚙️ **Power = rate at which work is done**

- For a body acted upon by force $\vec{F}$ and moving with velocity $\vec{v}$,



$$\text{Power} = \frac{dU}{dt} = \vec{F} \cdot \vec{v}$$



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Power

⚙️ **Power = rate at which work is done**

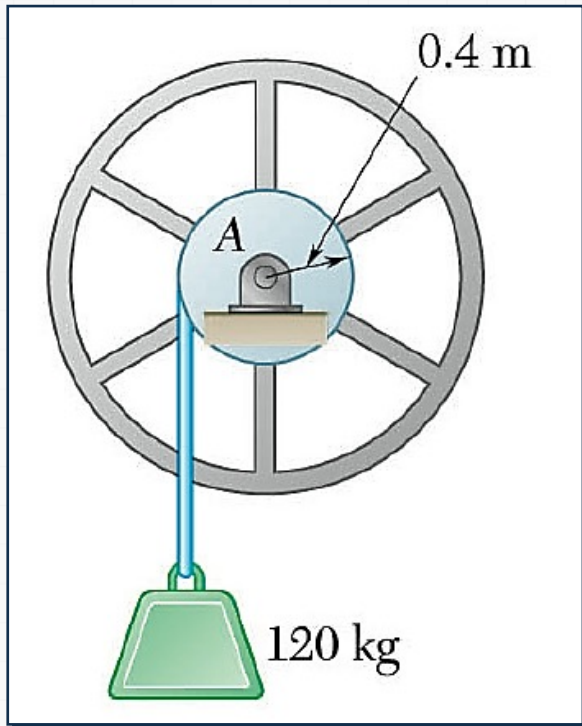
- For a rigid body rotating with an angular velocity $\vec{\omega}$ and acted upon by a couple of moment $\vec{M}$ parallel to the axis of rotation,

$$\text{Power} = \frac{dU}{dt} = \frac{M d\theta}{dt} = M\omega$$

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Sample Problem 1

⚙ For the drum and flywheel, $\bar{I} = 16 \text{ kgm}^2$.



- The bearing friction is equivalent to a couple of 90 Nm .
- At the instant shown, the block is moving downward at 2 m/s .
- Determine
 - Determine the velocity of the block after it has moved 1.25 m downward.

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Sample Problem 1

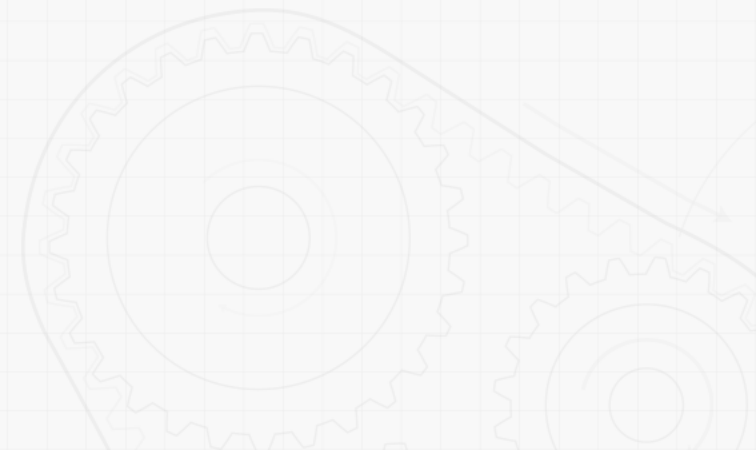
⚙️ **For the drum and flywheel, $\bar{I} = 16 \text{ kgm}^2$.**

• **Strategy**

- Consider the system of the flywheel and block. The work done by the internal forces exerted by the cable cancels.
- Note that the velocity of the block and the angular velocity of the drum and flywheel are related by

$$\bar{v} = r\omega$$

- Apply the principle of work and kinetic energy to develop an expression for the final velocity.

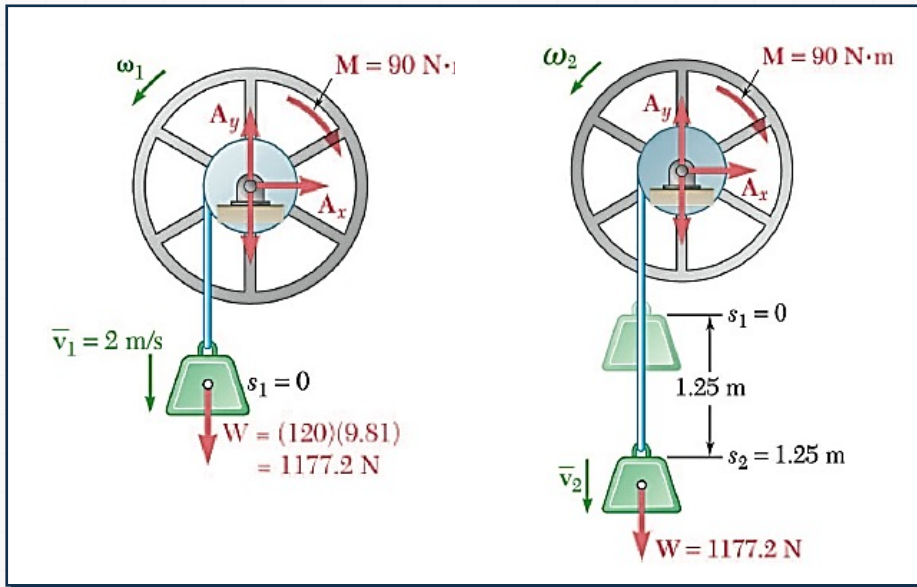


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Sample Problem 1

⚙ Modeling and Analysis

- Consider the system of the flywheel and block.



- The work done by the internal forces exerted by the cable cancels.
- Note that the velocity of the block and the angular velocity of the drum and flywheel are related by

$$\bar{v} = r\omega$$

$$\omega_1 = \frac{\bar{v}_1}{r} = \frac{2}{0.4} = 5 \text{ rad/s}$$

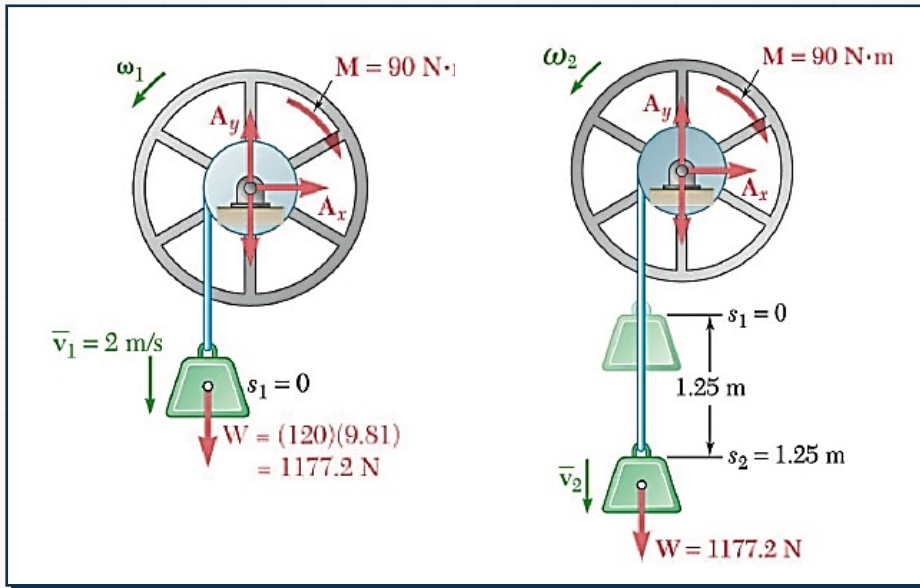
$$\omega_2 = \frac{\bar{v}_2}{r} = \frac{\bar{v}_2}{0.4}$$

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Sample Problem 1

⚙ Modeling and Analysis

- Consider the system of the flywheel and block.



- Apply the principle of work and kinetic energy to develop an expression for the final velocity.

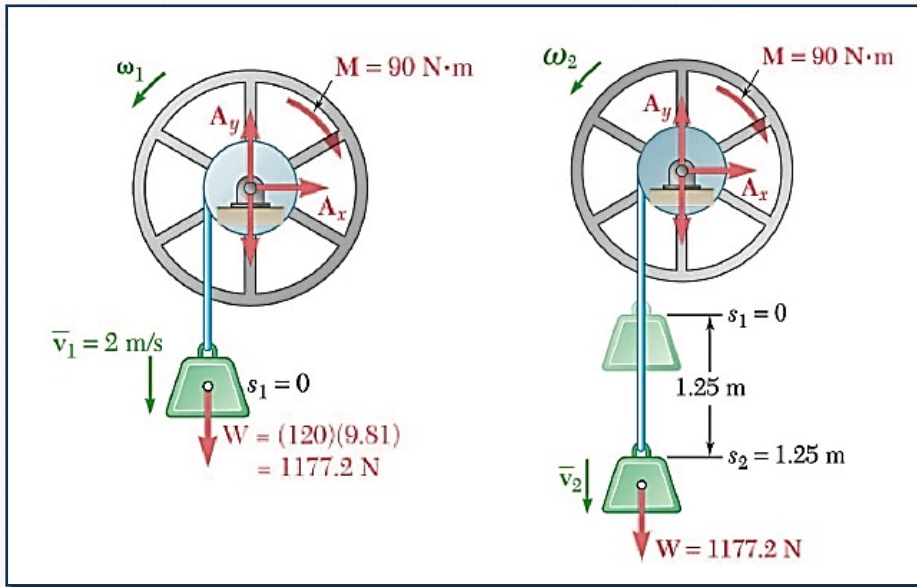
$$\begin{aligned}
 T_1 &= \frac{1}{2}mv_1^2 + \frac{1}{2}\bar{I}\omega_1^2 \\
 &= \frac{1}{2}(120)(2)^2 + \frac{1}{2}(16)(5)^2 \\
 &= 440 \text{ J}
 \end{aligned}$$

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Sample Problem 1

Modeling and Analysis

- Consider the system of the flywheel and block.



- Apply the principle of work and kinetic energy to develop an expression for the final velocity.

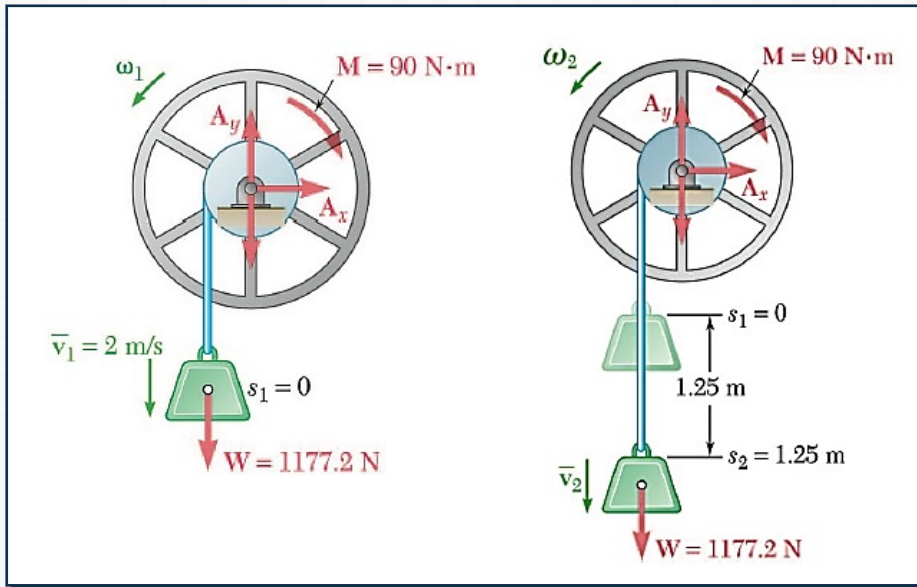
$$\begin{aligned}
 T_2 &= \frac{1}{2}mv_2^2 + \frac{1}{2}\bar{I}\omega_2^2 \\
 &= \frac{1}{2}(120)\bar{v}_2^2 + \frac{1}{2}(16)\left(\frac{\bar{v}_2}{0.4}\right)^2 = 110\bar{v}_2^2
 \end{aligned}$$

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Sample Problem 1

⚙ Modeling and Analysis

- Consider the system of the flywheel and block.



- Apply the principle of work and kinetic energy to develop an expression for the final velocity.

$$T_1 = \frac{1}{2}mv_1^2 + \frac{1}{2}I\omega_1^2 = 440\text{J}$$

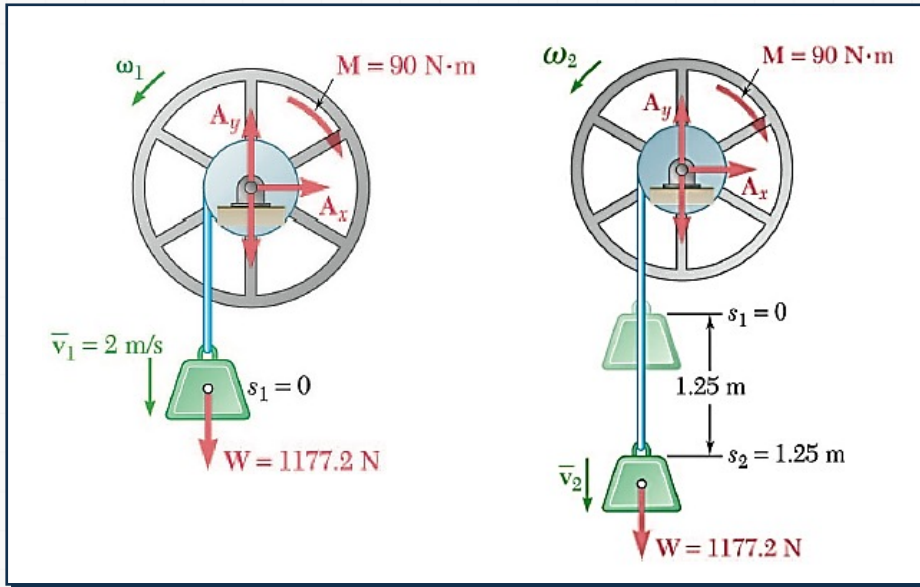
$$T_2 = \frac{1}{2}m\bar{v}_2^2 + \frac{1}{2}I\omega_2^2 = 110\bar{v}_2^2$$

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Sample Problem 1

⚙ Modeling and Analysis

- Consider the system of the flywheel and block.



- Note that the block displacement and pulley rotation are related by

$$\theta_2 = \frac{s_2}{r} = \frac{1.25}{0.4} = 3.125 \text{ rad}$$

Then,

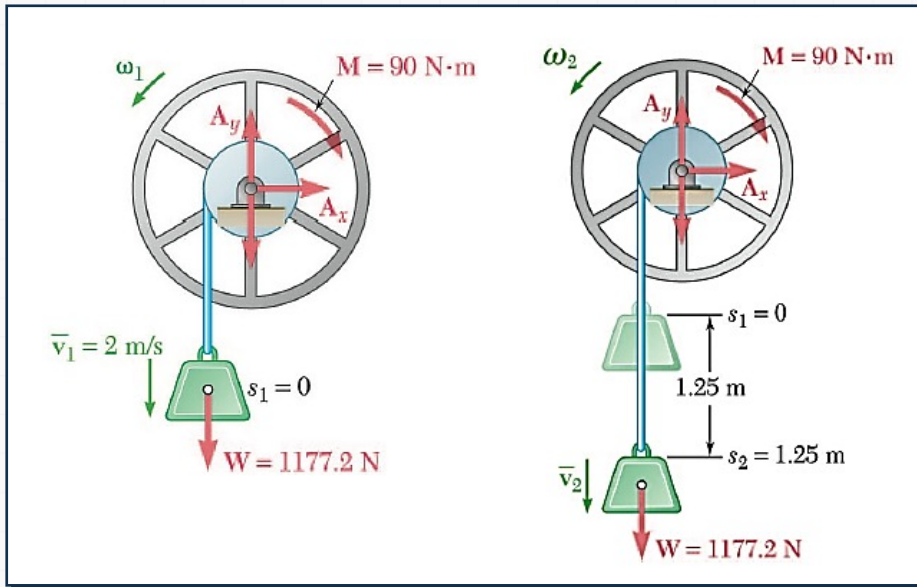
$$\begin{aligned} U_{1 \rightarrow 2} &= W(s_2 - s_1) - M(\theta_2 - \theta_1) \\ &= (1177.2)(1.25) - (90)(3.125) \\ &= 1190 \text{ J} \end{aligned}$$

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Sample Problem 1

⚙ Modeling and Analysis

- Consider the system of the flywheel and block.



- Principle of work and energy

$$T_1 + U_{1 \rightarrow 2} = T_2$$

$$440 + 1190 = 110\bar{v}_2^2$$

$$\bar{v}_2 = 3.85 \text{ m/s}$$

$$\bar{v}_2 = 3.85 \text{ m/s} \downarrow$$

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Sample Problem 1

⚙️ Reflect and Think

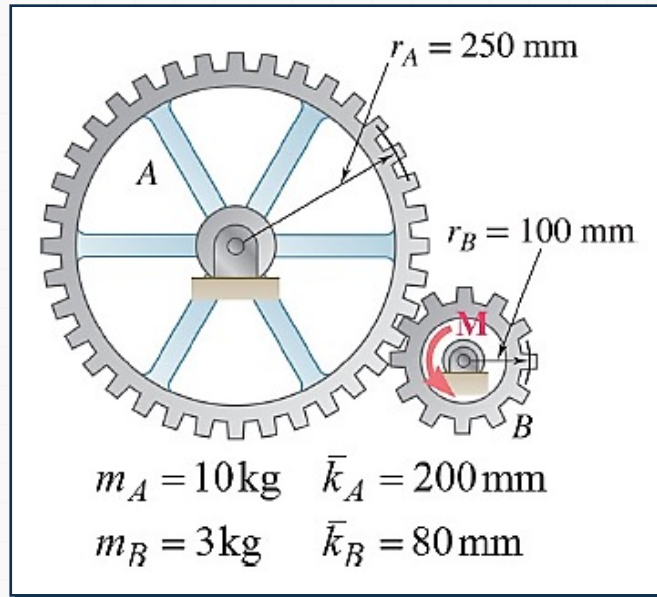
The speed of the block increases as it falls, but much more slowly than if it were in free fall. This seems like a reasonable result.

Rather than calculating the work done by gravity, you could have also treated the effect of the weight using gravitational potential energy, V_g .

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Sample Problem 2

⚙ The system is at rest when a moment of $M = 6 \text{ N} \cdot \text{m}$ is applied to gear B .



• Determine

- Neglecting friction,
 - determine the number of revolutions of gear B before its angular velocity reaches 600rpm
 - tangential force exerted by gear B on gear A .

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Sample Problem 2

⚙️ **The system is at rest when a moment of $M = 6 \text{ N} \cdot \text{m}$ is applied to gear B .**

• **Strategy**

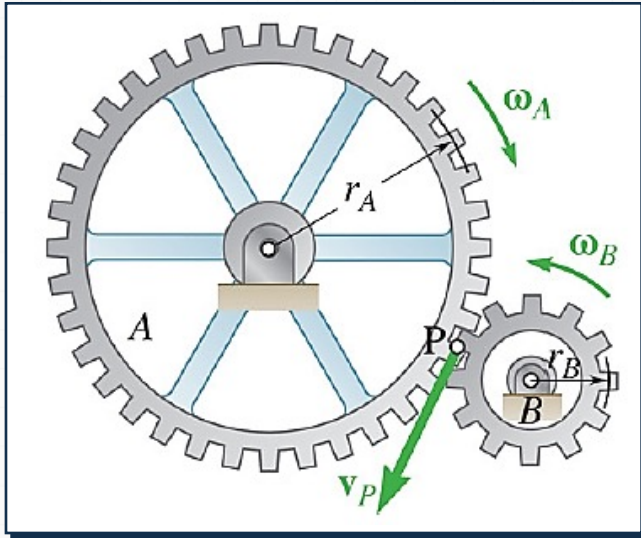
- Consider a system consisting of the two gears. Noting that the gear rotational speeds are related, evaluate the final kinetic energy of the system.
- Apply the principle of work and energy. Calculate the number of revolutions required for the work of the applied moment to equal the final kinetic energy of the system.
- Apply the principle of work and energy to a system consisting of gear A . With the final kinetic energy and number of revolutions known, calculate the moment and tangential force required for the indicated work.

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Sample Problem 2

⚙ Modeling and Analysis

- Consider a system consisting of the two gears.



- Noting that the gear rotational speeds are related, evaluate the final kinetic energy of the system.

$$\omega_B = \frac{(600)(2\pi)}{60} = 62.8 \text{ rad/s}$$

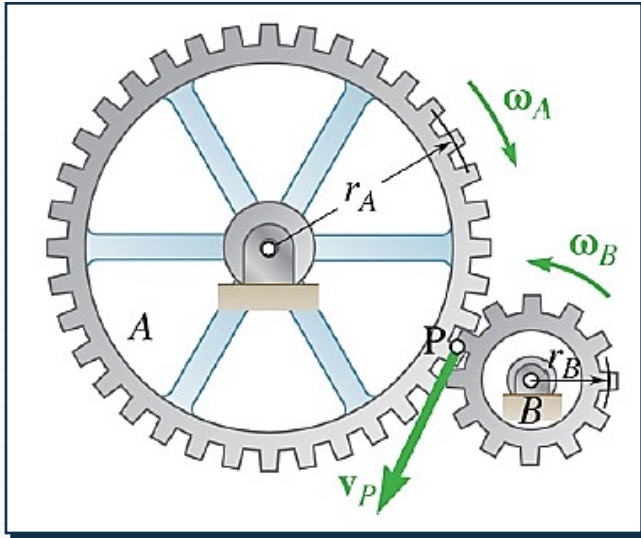
$$\omega_A = \omega_B \frac{r_B}{r_A} = 62.8 \frac{0.100}{0.250} = 25.1 \text{ rad/s}$$

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Sample Problem 2

⚙ Modeling and Analysis

- Consider a system consisting of the two gears.



- Noting that the gear rotational speeds are related, evaluate the final kinetic energy of the system.

$$\bar{I}_A = m_A \bar{k}_A^2 = (10)(0.200)^2 = 0.400 \text{ kg} \cdot \text{m}^2$$

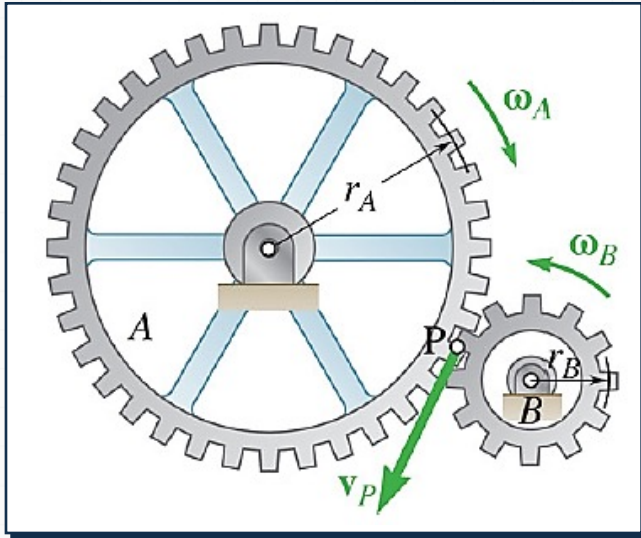
$$\bar{I}_B = m_B \bar{k}_B^2 = (3)(0.080)^2 = 0.0192 \text{ kg} \cdot \text{m}^2$$

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Sample Problem 2

⚙ Modeling and Analysis

- Consider a system consisting of the two gears.



- Noting that the gear rotational speeds are related, evaluate the final kinetic energy of the system.

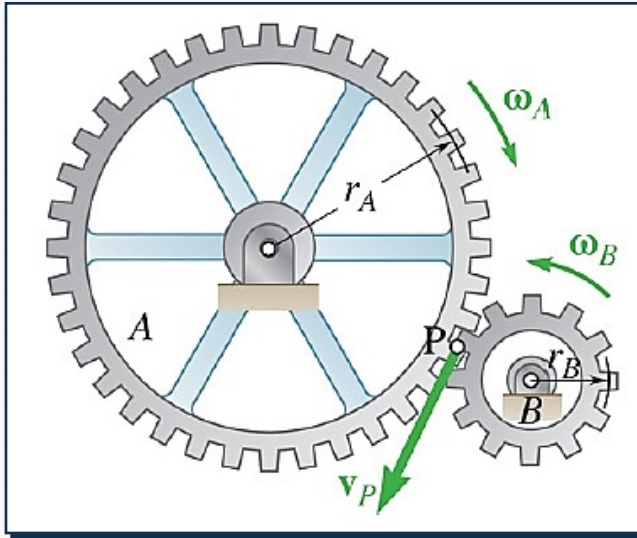
$$\begin{aligned} T_2 &= \frac{1}{2} \bar{I}_A \omega_A^2 + \frac{1}{2} \bar{I}_B \omega_B^2 \\ &= \frac{1}{2} (0.400) (25.1)^2 + \frac{1}{2} (0.0192) (62.8)^2 \\ &= 163.9 J \end{aligned}$$

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Sample Problem 2

⚙ Modeling and Analysis

- Consider a system consisting of the two gears.



- Apply the principle of work and energy. Calculate the number of revolutions required for the work.

$$T_1 + U_{1 \rightarrow 2} = T_2$$

$$0 + (6\theta_B) = 163.9\text{J}$$

$$\theta_B = 27.32\text{rad}$$

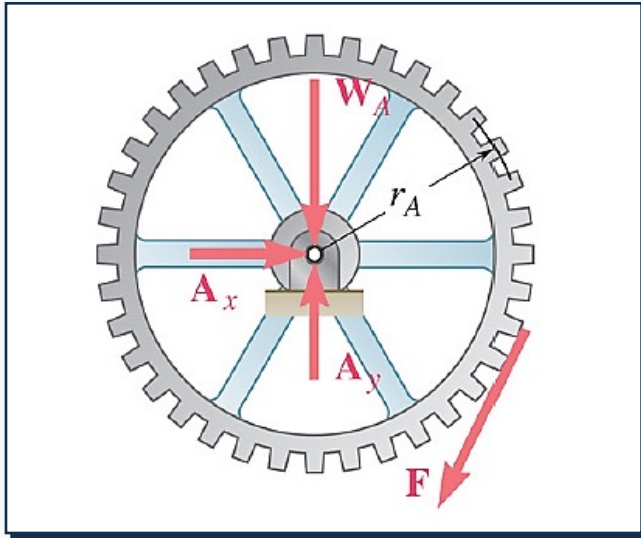
$$\theta_B = \frac{27.32}{2\pi} = 4.35\text{rev}$$

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Sample Problem 2

⚙ Modeling and Analysis

- Consider a system consisting of the two gears.



- Apply the principle of work and energy to a system consisting of gear A. Calculate the moment and tangential force required for the indicated work.

$$\theta_A = \theta_B \frac{r_B}{r_A} = 27.32 \frac{0.100}{0.250} = 10.93 \text{ rad}$$

$$T_2 = \frac{1}{2} \bar{I}_A \omega_A^2 = \frac{1}{2} (0.400) (25.1)^2 = 126.0 \text{ J}$$

$$T_1 + U_{1 \rightarrow 2} = T_2$$

$$0 + M_A (10.93) = 126.0 \text{ J}$$

$$M_A = r_A F = 11.52 \text{ N} \cdot \text{m}$$

$$F = \frac{11.52}{0.250} = 46.2 \text{ N}$$

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Sample Problem 2

⚙️ Reflect and Think

When the system was both gears, the tangential force between the gears did not appear in the work–energy equation, since it was internal to the system and therefore did no work.

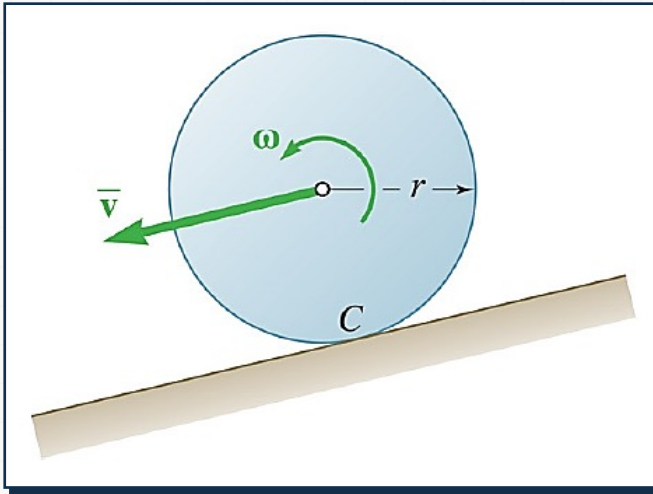
If you want to determine an internal force, you need to define a system where the force of interest is an external force.

This problem, like most problems, also could have been solved using Newton's second law and kinematic relationships.

14 Work and energy principle in planar motion

Sample Problem 3

⚙️ A sphere, cylinder, and hoop, each having the same mass and radius, are released from rest on an incline.



- Determine

- Determine the velocity of each body after it has rolled through a distance corresponding to a change of elevation h .

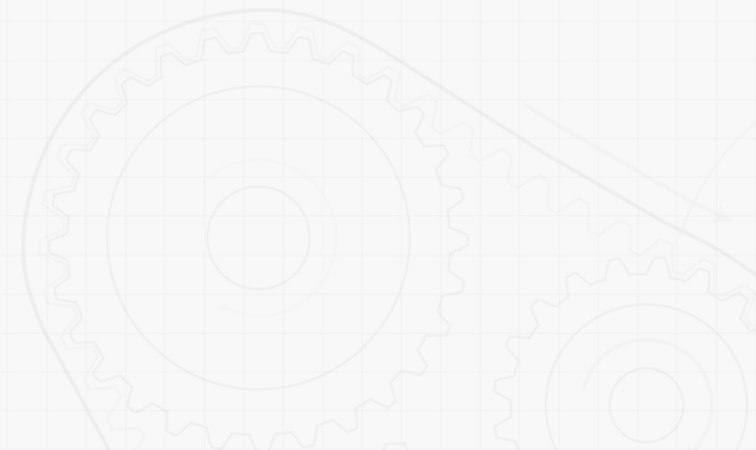
14 Work and energy principle in planar motion

Sample Problem 3

⚙️ **A sphere, cylinder, and hoop, each having the same mass and radius, are released from rest on an incline.**

- **Strategy**

- The work done by the weight of the bodies is the same. From the principle of work and energy, it follows that each body will have the same kinetic energy after the change of elevation.
- Because each of the bodies has a different centroidal moment of inertia, the distribution of the total kinetic energy between the linear and rotational components will be different as well.

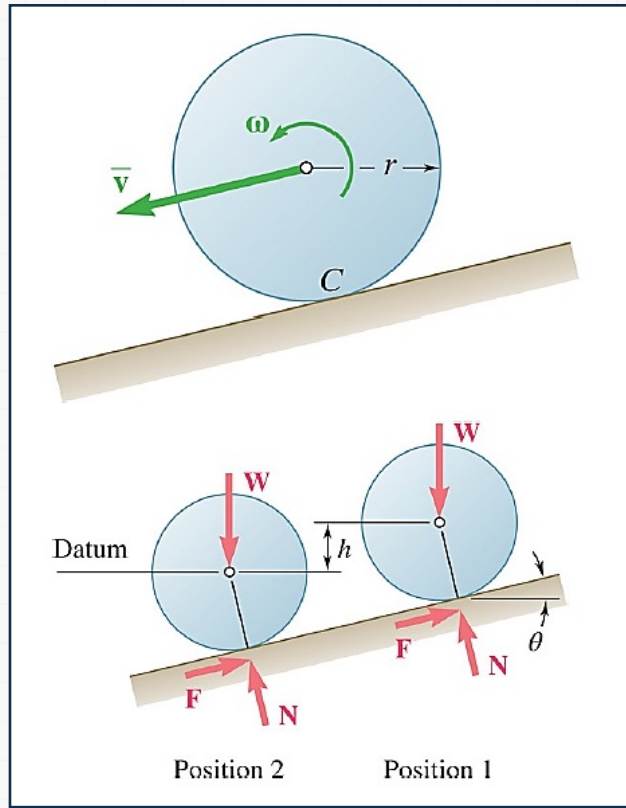


14 Work and energy principle in planar motion

Sample Problem 3

⚙ Modeling and Analysis

- The work done by the weight of the bodies is the same.



- From the principle of work and energy, it follows that each body will have the same kinetic energy after the change of elevation.

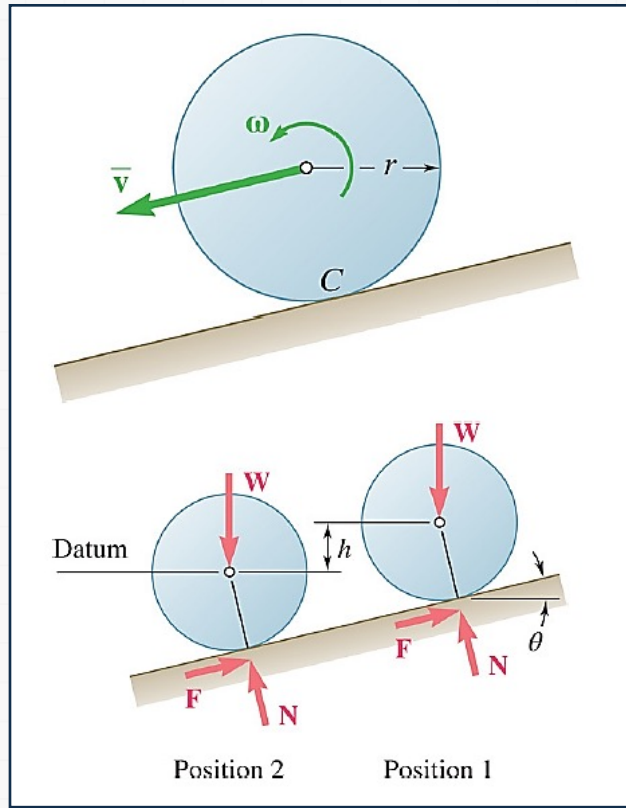
$$\begin{aligned}\text{With } \omega &= \frac{\bar{v}}{r} \\ T_2 &= \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \bar{I} \omega^2 = \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \bar{I} \left(\frac{\bar{v}}{r} \right)^2 \\ &= \frac{1}{2} \left(m + \frac{\bar{I}}{r^2} \right) \bar{v}^2\end{aligned}$$

14 Work and energy principle in planar motion

Sample Problem 3

⚙ Modeling and Analysis

- The work done by the weight of the bodies is the same.



- From the principle of work and energy, it follows that each body will have the same kinetic energy after the change of elevation.

$$T_1 + U_{1 \rightarrow 2} = T_2$$

$$0 + Wh = \frac{1}{2} \left(m + \frac{\bar{I}}{r^2} \right) \bar{v}^2$$

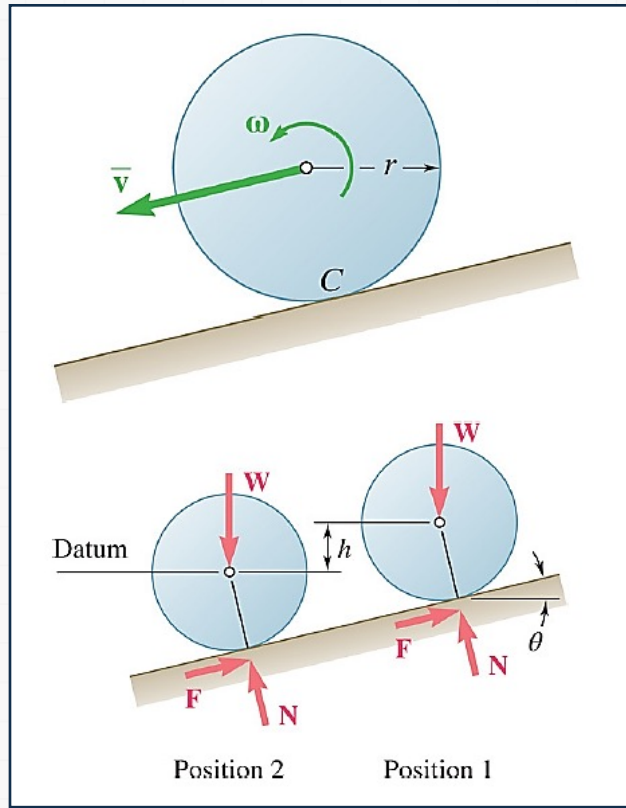
$$\bar{v}^2 = \frac{2Wh}{m + \bar{I}/r^2} = \frac{2gh}{1 + \bar{I}/mr^2}$$

14 Work and energy principle in planar motion

Sample Problem 3

⚙ Modeling and Analysis

- The distribution of the total kinetic energy



- Because each of the bodies has a different centroidal moment of inertia, the distribution of the total kinetic energy between the linear and rotational components will be different as well.

$$\bar{v}^2 = \frac{2gh}{1 + \bar{I}/mr^2}$$

Sphere $\bar{I} = \frac{2}{5}mr^2$ $\bar{v} = 0.845\sqrt{2gh}$

Cylinder $\bar{I} = \frac{1}{2}mr^2$ $\bar{v} = 0.816\sqrt{2gh}$

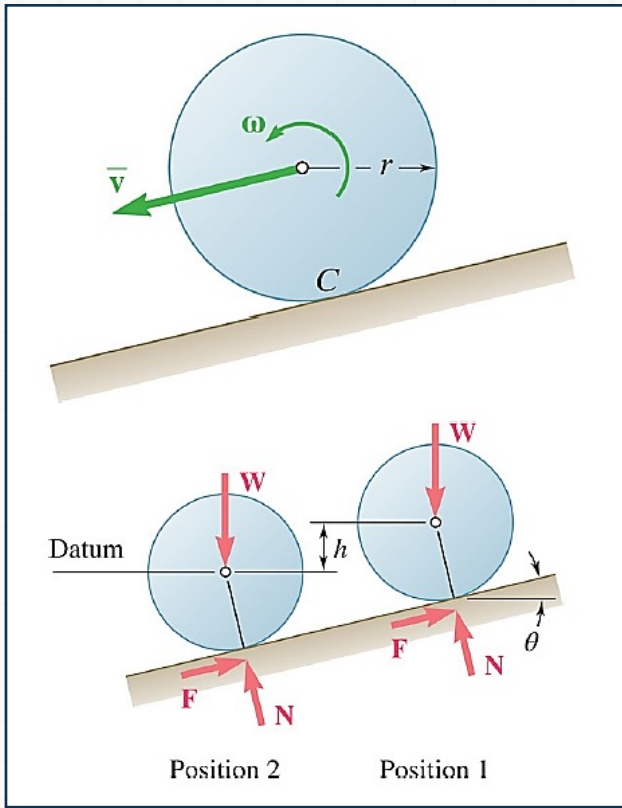
Hoop $\bar{I} = mr^2$ $\bar{v} = 0.707\sqrt{2gh}$

14 Work and energy principle in planar motion

Sample Problem 3

⚙ Modeling and Analysis

• NOTE



- For a frictionless block sliding through the same distance,

$$\omega = 0 \quad \bar{v} = \sqrt{2gh}$$

- The velocity of the body is independent of its mass and radius.
- The velocity of the body does depend on

$$\bar{I}/mr^2 = \bar{k}^2/r^2$$

14 Work and energy principle in planar motion

Sample Problem 3

⚙️ Reflect and Think

Let us compare the results with the velocity attained by a frictionless block sliding through the same distance. The solution is identical to the previous solution except that $\omega = 0$; we find $v = \sqrt{2gh}$.

14 Work and energy principle in planar motion

Sample Problem 3

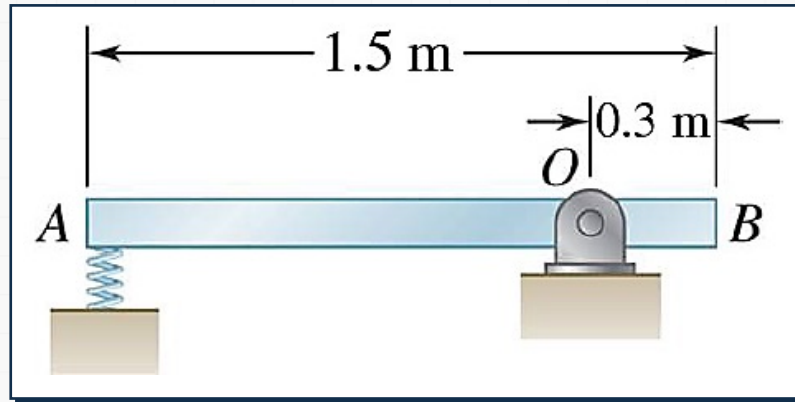
⚙️ Reflect and Think

Comparing the results, we note that the velocity of the body is independent of both its mass and radius. However, the velocity does depend upon the quotient of $I/mr^2 = k^2/r^2$, which measures the ratio of the rotational kinetic energy to the translational kinetic energy. Thus the hoop, which has the largest k for a given radius r , attains the smallest velocity, whereas the sliding block, which does not rotate, attains the largest velocity.

14 Work and energy principle in planar motion

Sample Problem 4

⚙️ A 15-kg slender rod pivots about the point O . The other end is pressed against a spring ($k = 300\text{ kN/m}$) until the spring is compressed 25 mm and the rod is in a horizontal position.



- Determine

- If the rod is released from this position, determine its angular velocity and the reaction at the pivot as the rod passes through a vertical position.

14 Work and energy principle in planar motion

Sample Problem 4

⚙️ **A 15-kg slender rod pivots about the point O . The other end is pressed against a spring ($k = 300\text{ kN/m}$) until the spring is compressed 25mm and the rod is in a horizontal position.**

• **Strategy**

- The weight and spring forces are conservative. The principle of work and energy can be expressed as

$$T_1 + V_1 = T_2 + V_2$$

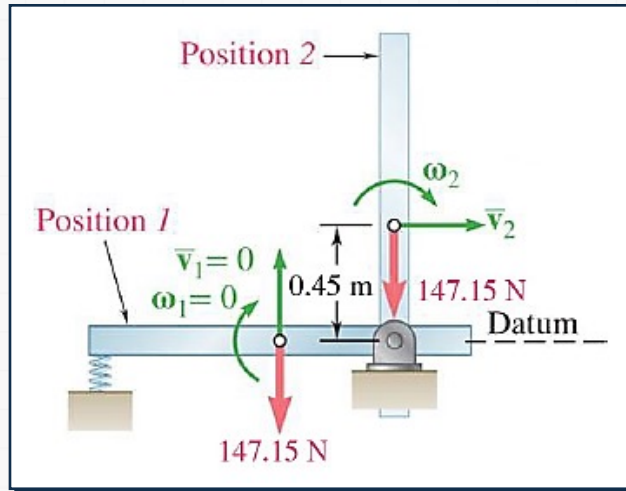
- Evaluate the initial and final potential energy.
- Express the final kinetic energy in terms of the final angular velocity of the rod.
- Based on the free-body-diagram equation, solve for the reactions at the pivot.

14 Work and energy principle in planar motion

Sample Problem 4

⚙ Modeling and Analysis

- The weight and spring forces are conservative.



- The principle of work and energy can be expressed as

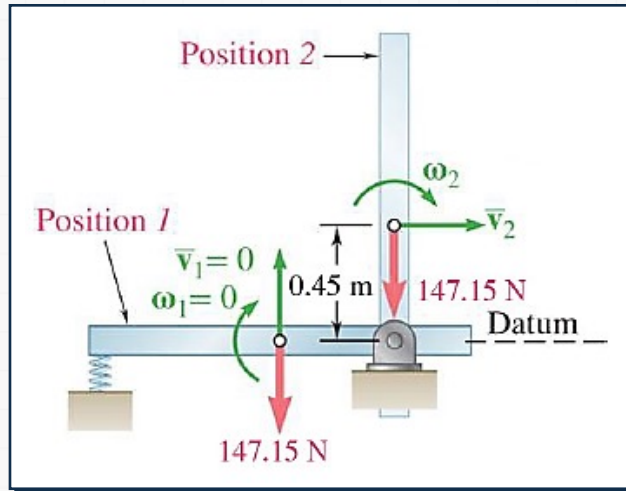
$$T_1 + V_1 = T_2 + V_2$$

14 Work and energy principle in planar motion

Sample Problem 4

⚙ Modeling and Analysis

- Evaluate the initial and final potential energy.



$$\begin{aligned} V_1 &= V_g + V_e = 0 + \frac{1}{2} kx_1^2 = \frac{1}{2} (300,000)(0.025)^2 \\ &= 93.75 \text{ Nm} \end{aligned}$$

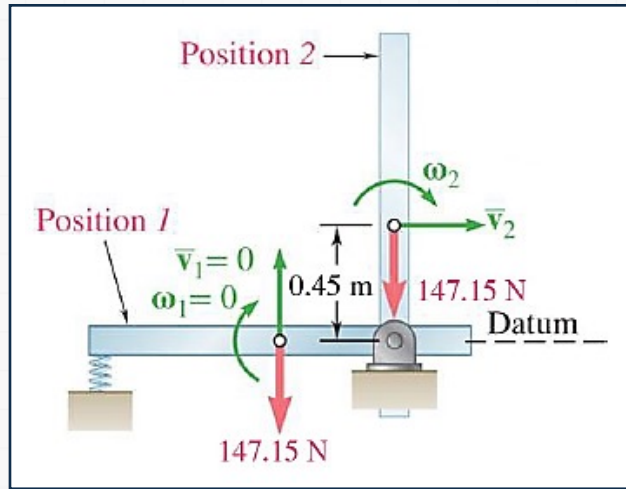
$$\begin{aligned} V_2 &= V_g + V_e = Wh + 0 = (147.15)(0.45) \\ &= 66.2175 \text{ J} \end{aligned}$$

14 Work and energy principle in planar motion

Sample Problem 4

⚙ Modeling and Analysis

- Express the final kinetic energy in terms of the angular velocity of the rod.



$$\bar{I} = \frac{1}{12}ml^2 = \frac{1}{12}(15)(1.5)^2 = 2.8125\text{kgm}^2$$

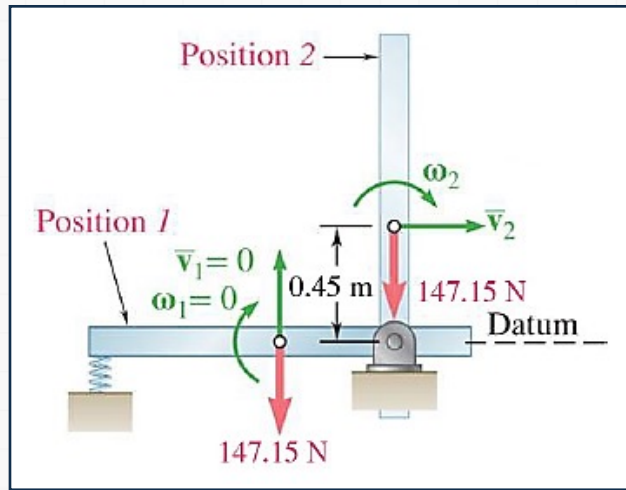
$$\begin{aligned} T_2 &= \frac{1}{2}m\bar{v}_2^2 + \frac{1}{2}\bar{I}\omega_2^2 = \frac{1}{2}m(r\omega_2)^2 + \frac{1}{2}\bar{I}\omega_2^2 \\ &= \frac{1}{2}(15)(0.45\omega_2)^2 + \frac{1}{2}(2.8125)\omega_2^2 = 2.925\omega_2^2 \end{aligned}$$

14 Work and energy principle in planar motion

Sample Problem 4

⚙ Modeling and Analysis

- Express the final kinetic energy in terms of the angular velocity of the rod.



- From the principle of work and energy,

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 93.75 = 2.925\omega_2^2 + 66.2175$$

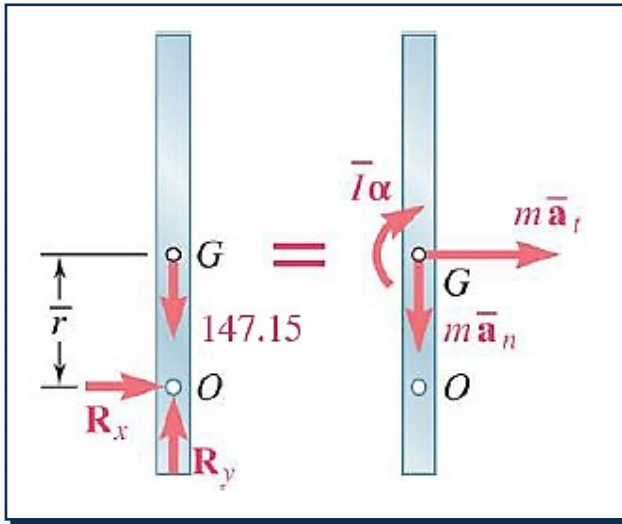
$$\omega_2 = 3.068 \text{ rad/s} \curvearrowright$$

14 Work and energy principle in planar motion

Sample Problem 4

⚙ Modeling and Analysis

- Based on the free-body-diagram equation, solve for the reactions at the pivot.



$$\bar{a}_n = \bar{r}\omega_2^2 = (0.45)(3.068)^2 = 4.236\text{m/s}^2$$

$$\vec{\bar{a}}_n = 4.236\text{m/s}^2 \downarrow$$

$$\bar{a}_t = r\alpha$$

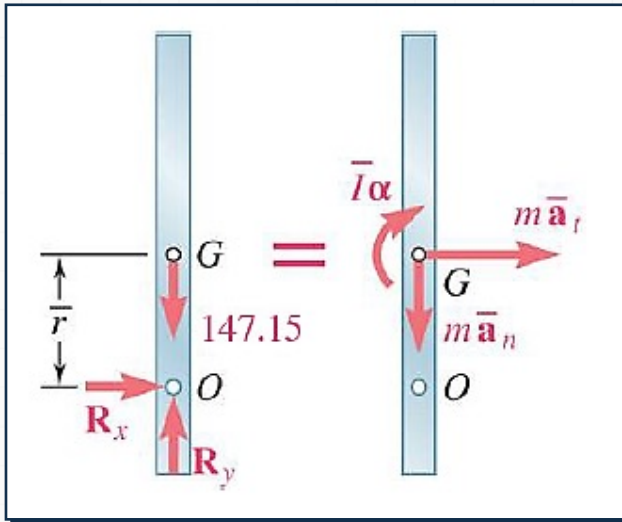
$$\vec{\bar{a}}_t = r\alpha \rightarrow$$

14 Work and energy principle in planar motion

Sample Problem 4

⚙ Modeling and Analysis

- Based on the free-body-diagram equation, solve for the reactions at the pivot.



$$+\curvearrowright \sum M_O = \sum (M_O)_{eff} \quad 0 = \bar{I}\alpha + m(\bar{r}\alpha)\bar{r} \quad \alpha = 0$$

$$+\rightarrow \sum F_x = \sum (F_x)_{eff} \quad R_x = m(\bar{r}\alpha) \quad R_x = 0$$

$$+\uparrow \sum F_y = \sum (F_y)_{eff} \quad R_y - 147.15\text{N} = -ma_n$$

$$= -(15)(4.236)$$

$$R_y = 83.61\text{N}$$

$$\vec{R} = 83.6\text{ N } \uparrow$$

14 Work and energy principle in planar motion

Sample Problem 4

⚙️ Reflect and Think

This problem illustrates how you might need to supplement the conservation of energy with Newton's second law.

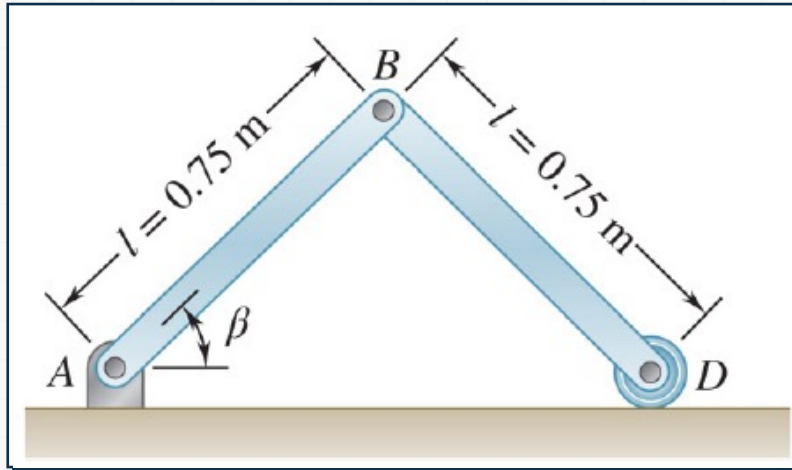
What if the spring constant had been smaller, say 30 kN/m ? You would have found $V_{e1} = 9.375 \text{ J}$ and then solved to obtain $\omega_2^2 = -19.43$.

This is clearly impossible and means that the rod would not make it to position 2 as assumed.

14 Work and energy principle in planar motion

Sample Problem 5

⚙ Each of the two slender rods has a mass of 6 kg .
The system is released from rest with $\beta = 60^\circ$.



• Determine

- a) the angular velocity of rod AB when $\beta = 20^\circ$
- b) the velocity of the point D at the same instant.

14 Work and energy principle in planar motion

Sample Problem 5

⚙️ **Each of the two slender rods has a mass of 6 kg .
The system is released from rest with $\beta = 60^\circ$.**

• **Strategy**

- Consider a system consisting of the two rods.
With the conservative weight force,

$$T_1 + V_1 = T_2 + V_2$$

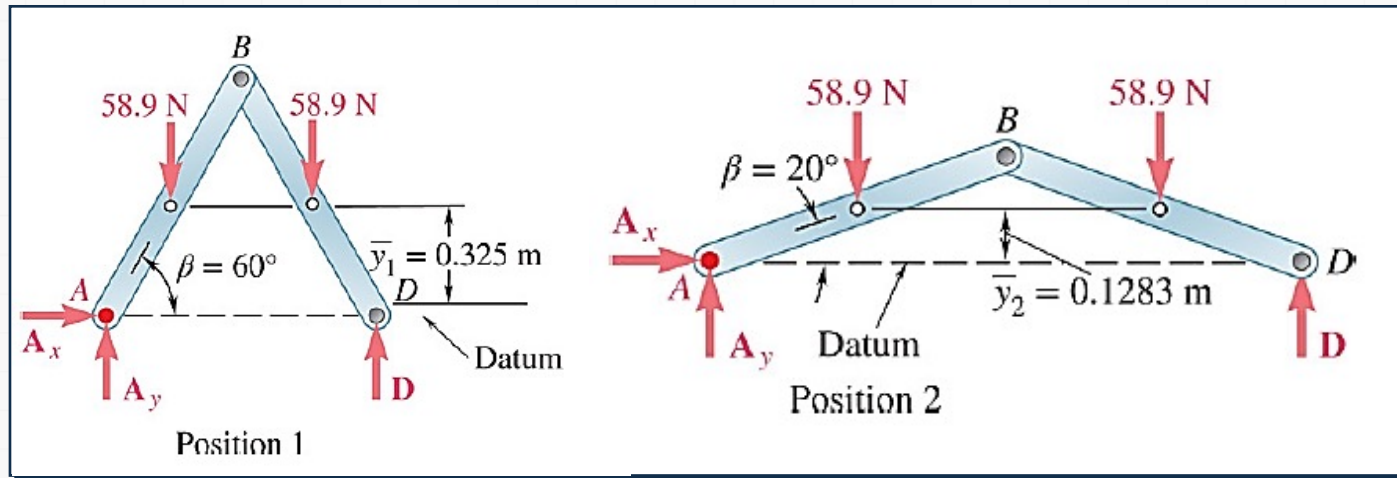
- Evaluate the initial and final potential energy.
- Express the final kinetic energy of the system in terms of the angular velocities of the rods.
- Solve the energy equation for the angular velocity, then evaluate the velocity of the point D .

14 Work and energy principle in planar motion

Sample Problem 5

Modeling and Analysis

- Consider a system consisting of the two rods.



- With the conservative weight force,

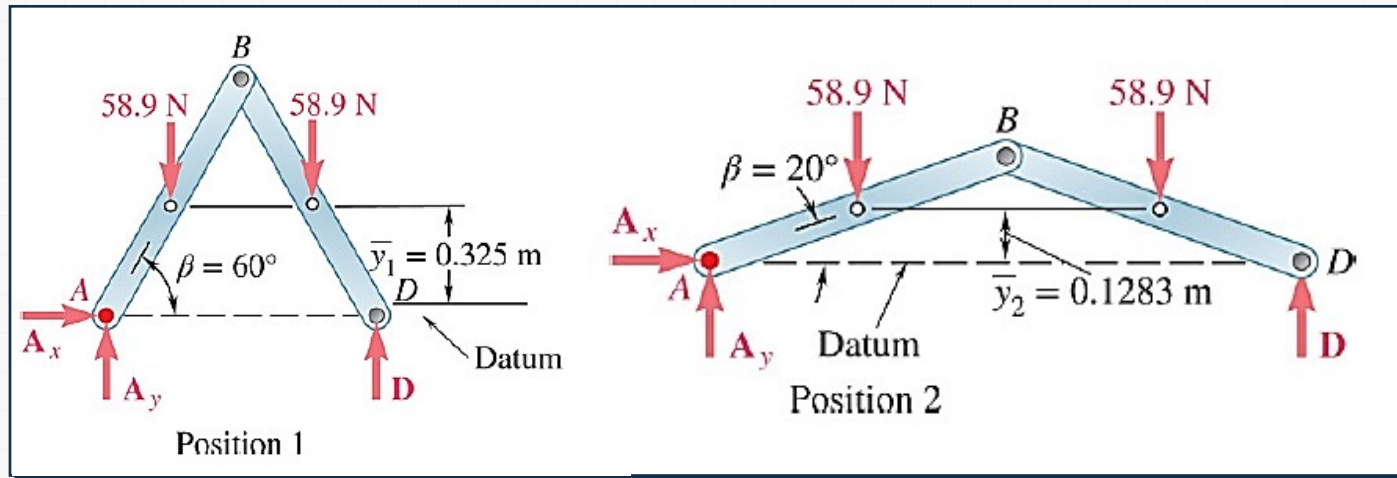
$$T_1 + V_1 = T_2 + V_2$$

14 Work and energy principle in planar motion

Sample Problem 5

Modeling and Analysis

- Evaluate the initial and final potential energy.



$$W = mg = (6)(9.81) = 58.86 \text{ N}$$

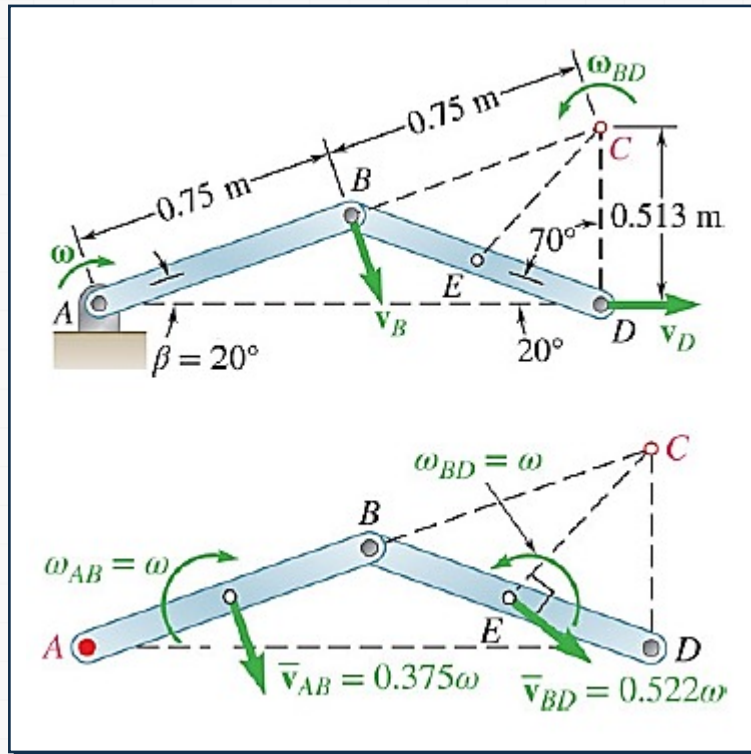
$$V_1 = 2W\bar{y}_1 = 2(58.86)(0.325) = 38.26 \text{ J}$$

$$V_2 = 2W\bar{y}_2 = 2(58.86)(0.1283) = 15.10 \text{ J}$$

14 Work and energy principle in planar motion

Sample Problem 5

⚙ Modeling and Analysis



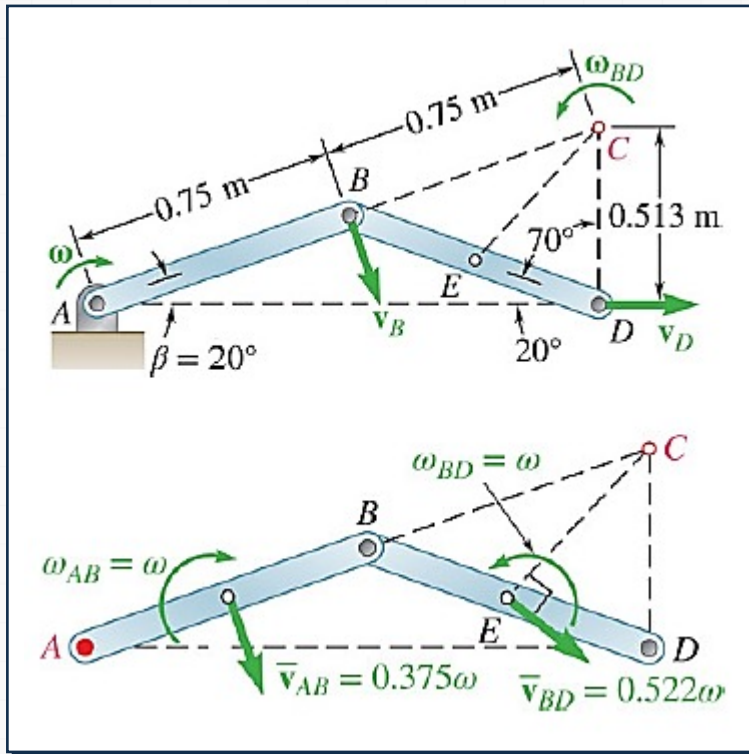
- Express the final kinetic energy of the system in terms of the angular velocities of the rods.

$$v_{AB} = (0.375)\omega$$

14 Work and energy principle in planar motion

Sample Problem 5

Modeling and Analysis



- Express the final kinetic energy of the system in terms of the angular velocities of the rods.

- Since $\vec{v}_B$ is perpendicular to AB and $\vec{v}_D$ is horizontal, the instantaneous center of rotation for rod BD is C.

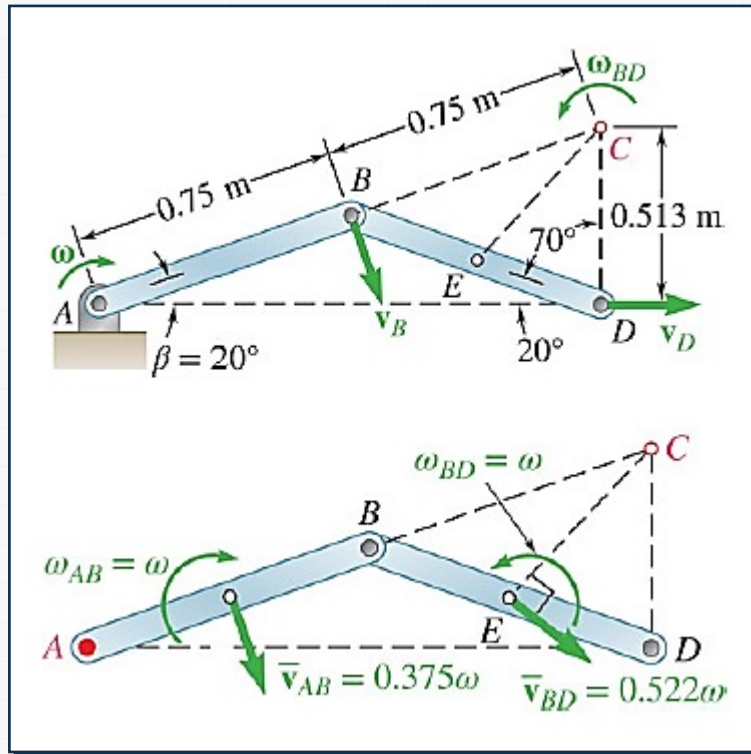
$$BC = 0.75\text{m} \quad CD = 2(0.75)\sin 20^\circ = 0.513\text{m}$$

and applying the law of cosines to CDE , $EC = 0.522\text{ m}$

14 Work and energy principle in planar motion

Sample Problem 5

Modeling and Analysis



- Express the final kinetic energy of the system in terms of the angular velocities of the rods.
 - Consider the velocity of point B

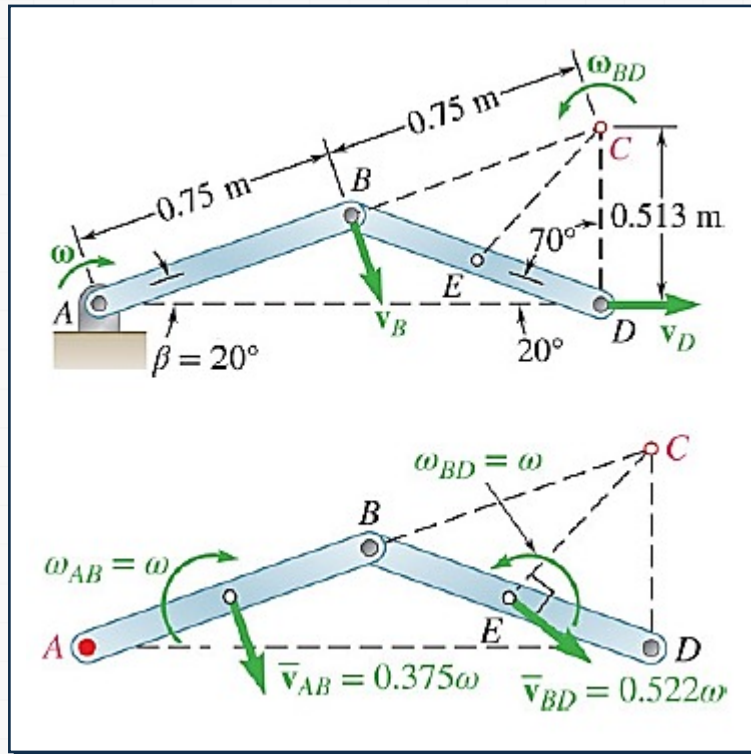
$$v_B = (AB)\omega = (BC)\omega_{BD} \quad \vec{\omega}_{BD} = \omega \curvearrowright$$

$$\vec{v}_{BD} = (0.522\text{m})\omega \searrow$$

14 Work and energy principle in planar motion

Sample Problem 5

⚙️ Modeling and Analysis



• For the final kinetic energy,

$$\bar{I}_{AB} = \bar{I}_{BD} = \frac{1}{12} m l^2 = \frac{1}{12} (6) (0.75)^2 = 0.281 \text{ kg} \cdot \text{m}^2$$

$$T_2 = \frac{1}{12} m \bar{v}_{AB}^2 + \frac{1}{2} \bar{I}_{AB} \omega_{AB}^2 + \frac{1}{12} m \bar{v}_{BD}^2 + \frac{1}{2} \bar{I}_{BD} \omega_{BD}^2$$

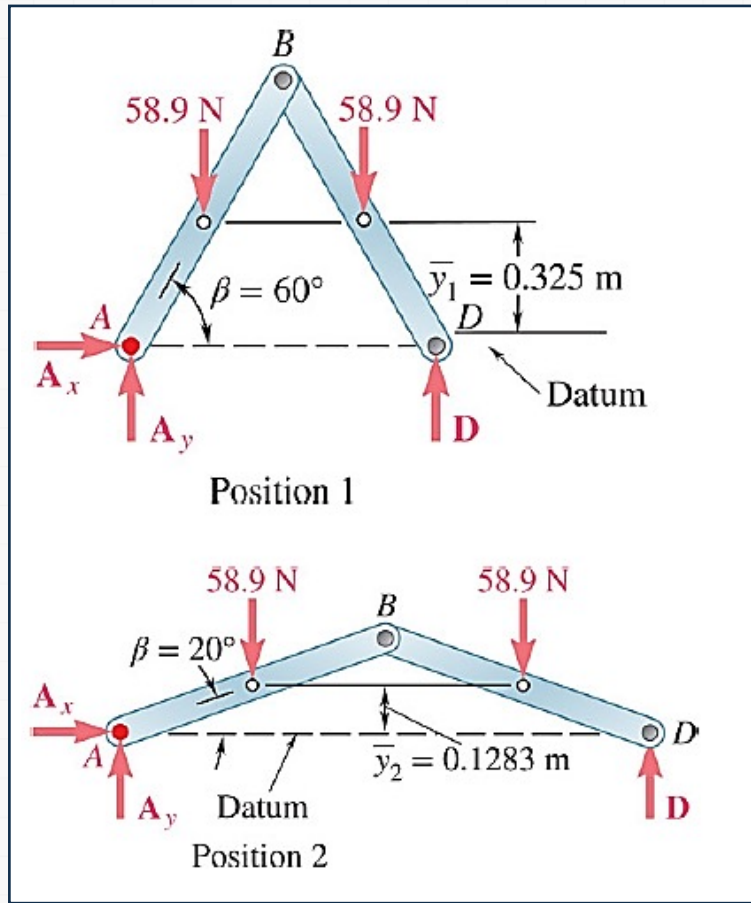
$$= \frac{1}{12} (6) (0.375\omega)^2 + \frac{1}{2} (0.281) \omega^2 + \frac{1}{12} (6) (0.522\omega)^2 + \frac{1}{2} (0.281) \omega^2$$

$$= 1.520 \omega^2$$

14 Work and energy principle in planar motion

Sample Problem 5

Modeling and Analysis



- Solve the energy equation for the angular velocity, then evaluate the velocity of the point D .

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 38.26 = 1.520\omega^2 + 15.10$$

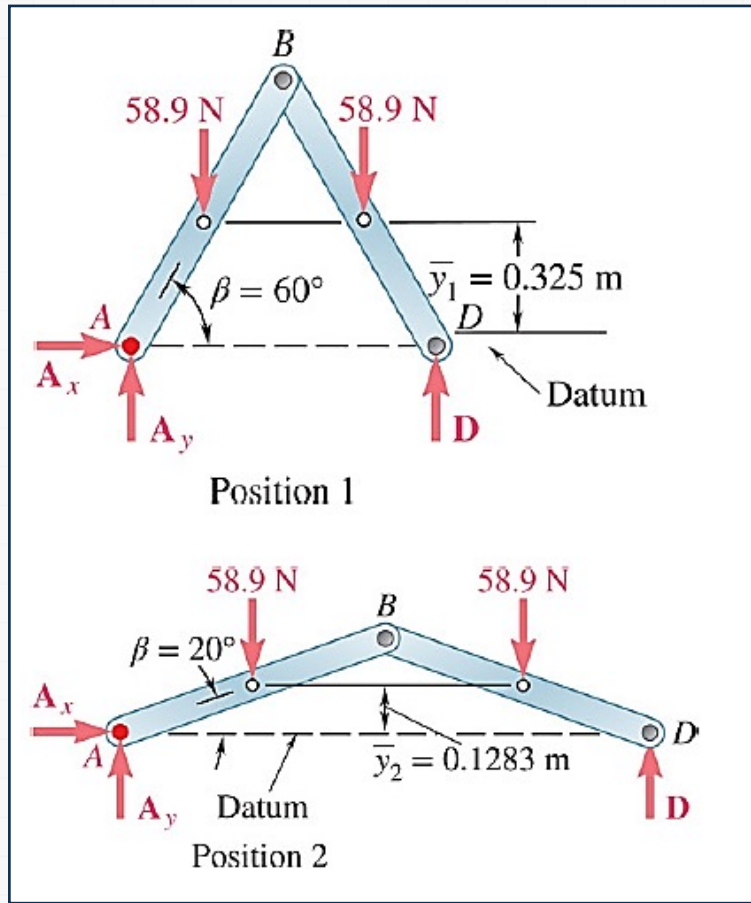
$$\omega = 3.90 \text{ rad/s}$$

$$\vec{\omega}_{AB} = 3.90 \text{ rad/s} \curvearrowright$$

14 Work and energy principle in planar motion

Sample Problem 5

Modeling and Analysis



- Solve the energy equation for the angular velocity, then evaluate the velocity of the point D .

$$T_1 + V_1 = T_2 + V_2$$

$$\begin{aligned} v_D &= (CD)\omega \\ &= (0.513)(3.90) \\ &= 2.00\text{m/s} \end{aligned}$$

$$\vec{v}_D = 2.00\text{m/s} \rightarrow$$

14 Work and energy principle in planar motion

Sample Problem 5

⚙️ Reflect and Think

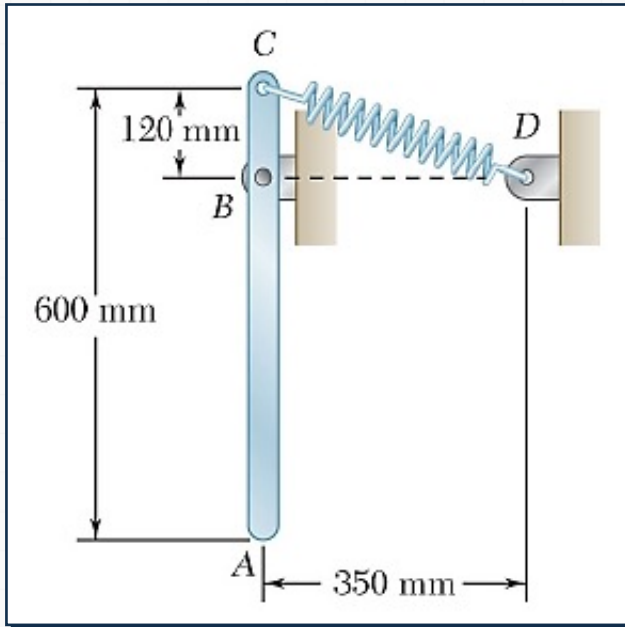
The only step in which you need to use forces is when calculating the gravitational potential energy in each position.

However, it is good engineering practice to show the complete free-body diagram in each case to identify which, if any, forces do work. You could have also used vector algebra to relate the velocities of the various objects.

14 Work and energy principle in planar motion

Group Problem Solving

⚙️ A slender $4 - kg$ rod can rotate in a vertical plane about a pivot at B .



• Determine

- A spring of constant $k = 400\text{ N/m}$ and of unstretched length 150 mm is attached to the rod as shown.
- Knowing that the rod is released from rest in the position shown, determine its angular velocity after it has rotated through 90° .

14 Work and energy principle in planar motion

Group Problem Solving

⚙️ **A slender $4 - kg$ rod can rotate in a vertical plane about a pivot at B .**

• Strategy

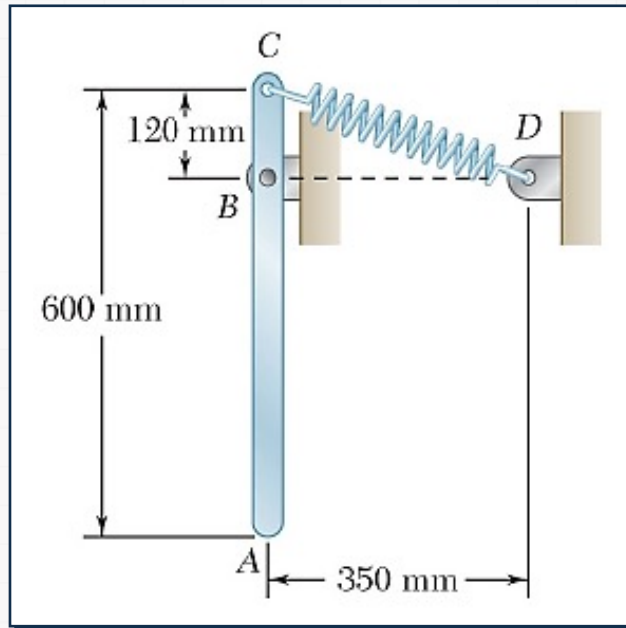
- Because the problem deals with positions and velocities, you should apply the principle of work energy.
- Draw out the system at position 1 and position 2 and define your datum.
- Use the work-energy equation to determine the angular velocity at position 2.

14 Work and energy principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis

- Draw your diagrams, set your datum and apply the work energy equation.



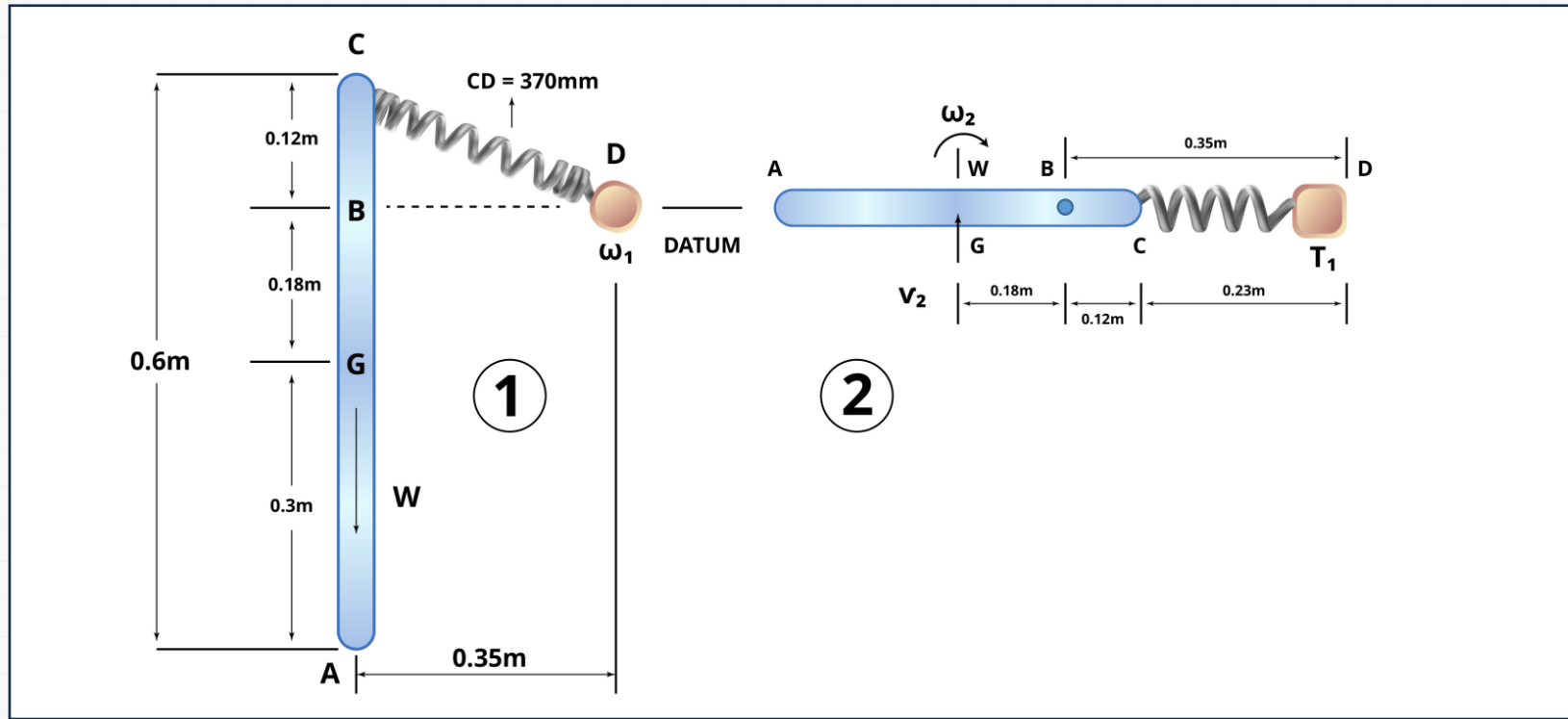
$$T_1 + V_1 + U_{1-2} = T_2 + V_2$$

14 Work and energy principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis

- Draw your diagrams, set your datum and apply the work energy equation.



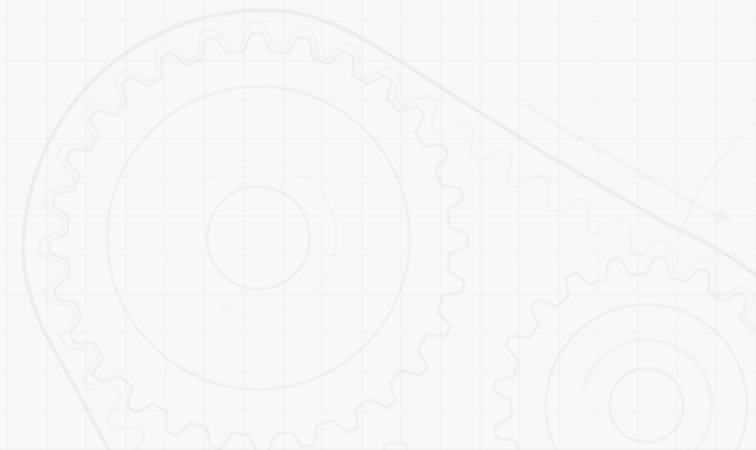
14 Work and energy principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis

- Are any of the terms zero?

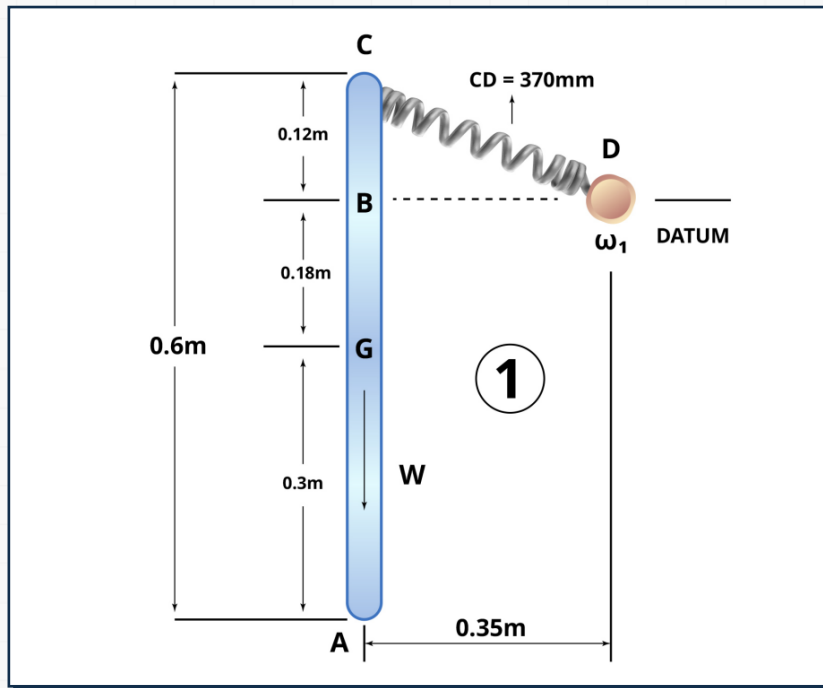
$$\cancel{T_1} + V_1 + \cancel{U_{1-2}} = T_2 + V_2$$



14 Work and energy principle in planar motion

Group Problem Solving

⚙️ Modeling and Analysis



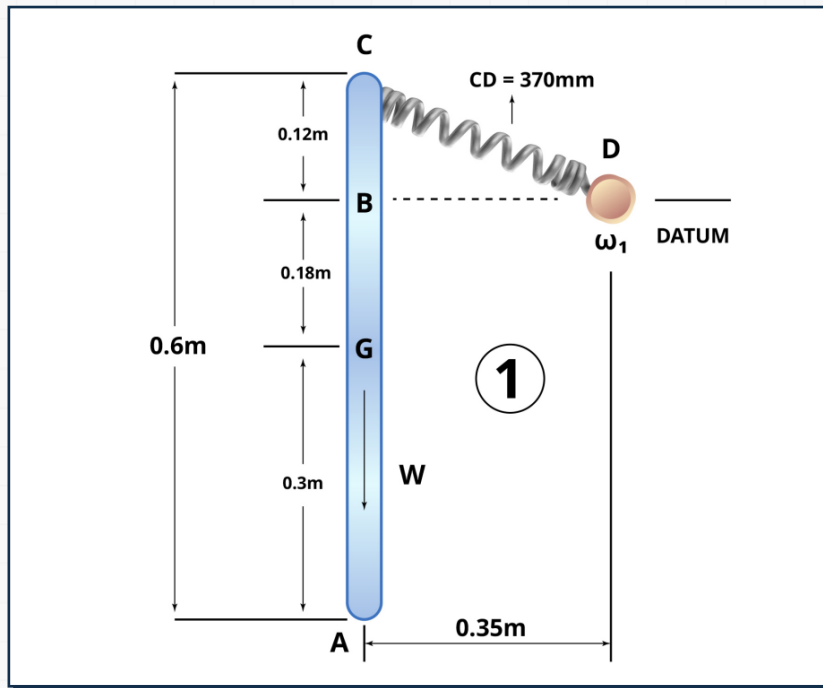
- Determine the spring energy at position 1

$$\begin{aligned} x_1 &= CD - \overbrace{(150\text{mm})}^{\text{Unstretched Length}} \\ &= 370 - 150 = 220\text{mm} = 0.22\text{m} \\ V_e &= \frac{1}{2} k x_1^2 = \frac{1}{2} (400)(0.22)^2 = 9.68\text{J} \end{aligned}$$

14 Work and energy principle in planar motion

Group Problem Solving

⚙️ Modeling and Analysis



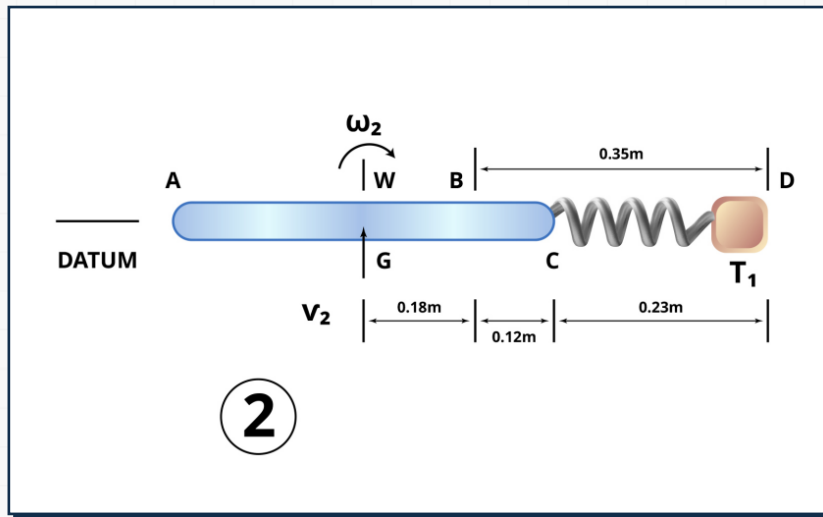
- Determine the potential energy due to gravity at position 1

$$\begin{aligned} V_{g1} &= Wh = mgh = (4)(9.81)(-0.22) \\ &= -7.063\text{J} \end{aligned}$$

14 Work and energy principle in planar motion

Group Problem Solving

⚙️ Modeling and Analysis



- Determine the spring energy at position 2

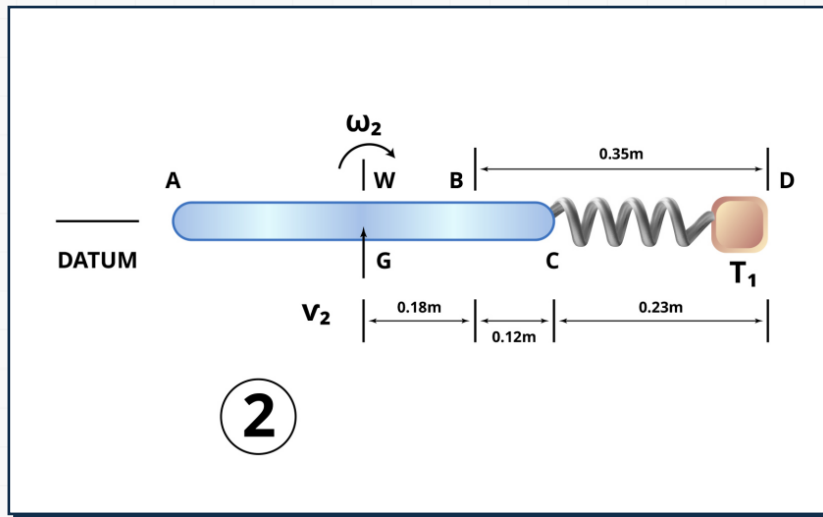
$$x_2 = 230 - 150 = 80\text{mm} = 0.08\text{m}$$

$$V_{e2} = \frac{1}{2} k x_2^2 = \frac{1}{2} (400) (0.08)^2 = 1.28\text{J}$$

14 Work and energy principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



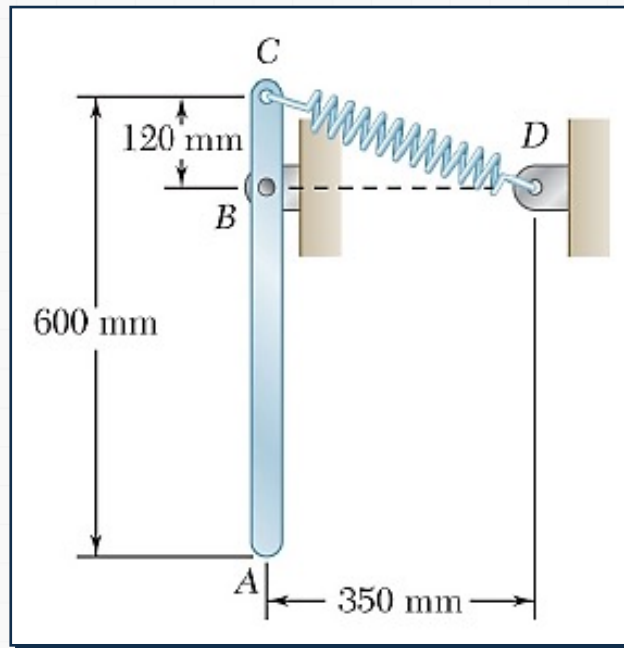
- Determine the potential energy due to gravity at position 2

$$V_{g2} = 0$$

14 Work and energy principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



- Determine an expression for T_2 .

$$T_2 = \frac{1}{2} m \bar{v}_2^2 + \frac{1}{2} \bar{I} \omega_2^2$$

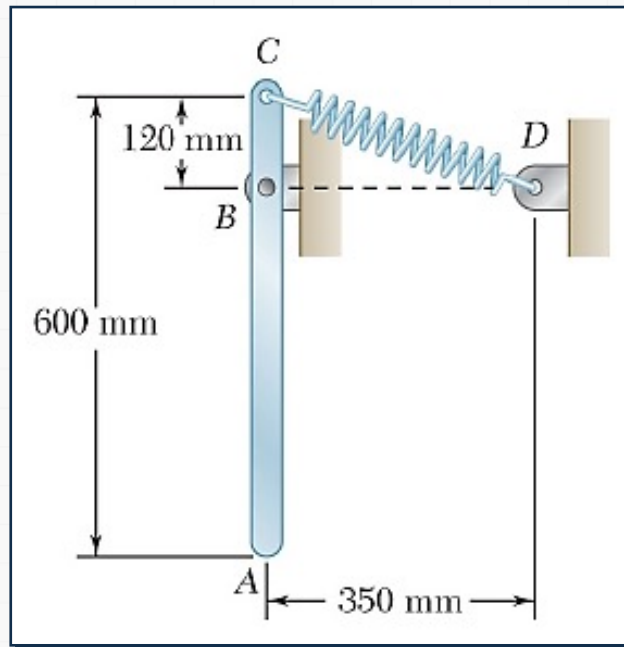
- Can you relate v_2 and ω_2 ?

$$\bar{v}_2 = r \omega_2 = (0.18) \omega_2$$

14 Work and energy principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



- Find I and substitute in to T_2 .

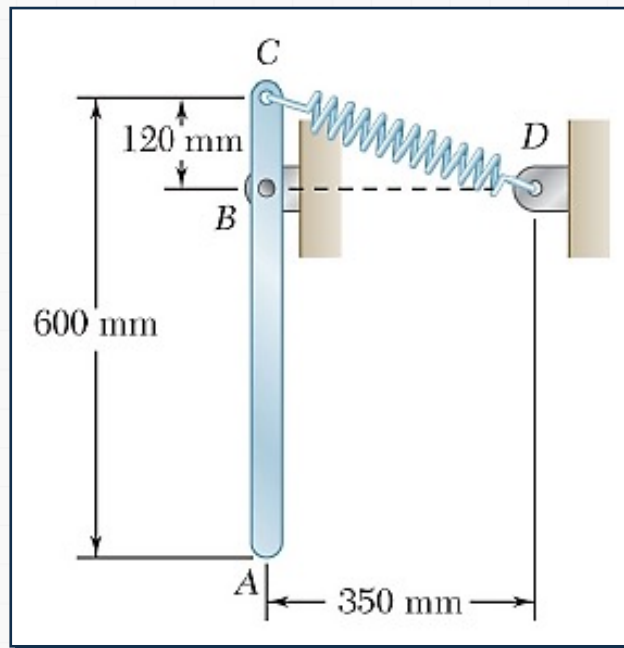
$$\bar{I} = \frac{1}{12} mL^2 = \frac{1}{12} (4)(0.6)^2 = 0.12 \text{ kg} \cdot \text{m}^2$$

$$T_2 = \frac{1}{2} m \bar{v}_2^2 + \frac{1}{2} \bar{I} \omega_2^2 = \frac{1}{2} (4)(0.18\omega_2)^2 + \frac{1}{2} (0.12)\omega_2^2 = 0.1248\omega_2^2$$

14 Work and energy principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



- Substitute into $T_1 + V_1 = T_2 + V_2$

$$9.68 - 7.063 = 0.1248\omega_2^2 + 1.28$$

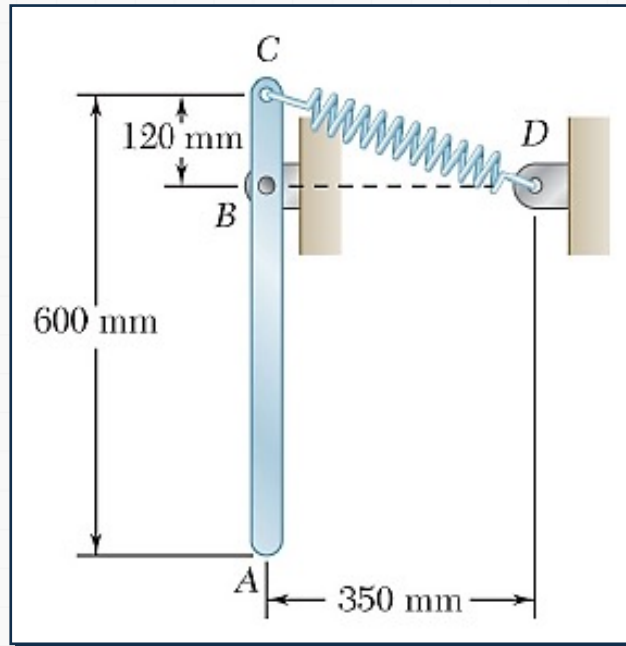
$$\omega_2^2 = 10.713$$

$$\omega_2 = 3.27 \text{ rad/s}$$

14 Work and energy principle in planar motion

Concept Question

⚙ For the previous problem, how would you determine the reaction forces at B when the bar is horizontal?



- a) Apply linear-momentum to solve for $B_x \Delta t$ and $B_y \Delta t$.
- b) Use work-energy to determine the work done by the moment at C .
- c) Use sum of forces and sum of moments equations when the bar is horizontal.

14 Work and energy principle in planar motion

| References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, Chapter 17